MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

BOOTSTRAP MEMORY/RTC MODULE, KEY CODE CWAF

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic bootstrap memory/Real Time Clock (RTC) module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 2.667 failures per million operating hours for class I or 6.773 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be CWAF.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B7	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	---------Module shall pass all remaining tests---------						
D1-D10	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D11	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D12	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D13-D24	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D25-D28	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D29	Ibd	V	S	0.0	3.1	0.0	3.1	0.0	3.1	mA
D30	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D31	Ibd	V	S	0.0	3.4	0.0	3.4	0.0	3.4	mA
D32	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D33-D43	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D44	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D45-D50	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
E1-E8	Ios	VI	S	-85.0	-12.0	-85.0	-12.0	-85.0	-12.0	mA
E9-E24	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E25	Ios	VI	S	-100.0	-15.0	-100.0	-15.0	-100.0	-15.0	mA
E26-E33	Ios	VI	S	-85.0	-12.0	-85.0	-12.0	-85.0	-12.0	mA
E34,E35	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E36,E37	Ios	VI	S	-100.0	-15.0	-100.0	-15.0	-100.0	-15.0	mA
E38	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
F1-F6	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F7-F22	Iih	VII	X	-0.22	0.04	-0.22	0.04	-0.22	0.04	mA
F23-F25	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ∓ 5°C		Class II 4/ -55 ± 5°C 85 ∓ 5°C		Units
				Min	Max	Min	Max	Min	Max	
F26	IIh	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F27	IIh	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F28	IIh	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F29-F40	IIh	VII	X	-0.5	0.02	-0.5	0.02	-0.5	0.02	mA
F41-F44	IIh	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F45	IIh	VII	X	-2.7	0.09	-2.7	0.09	-2.7	0.09	mA
F46	IIh	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F47	IIh	VII	X	-8.4	0.35	-8.4	0.35	-8.4	0.35	mA
F48	IIh	VII	X	-4.0	0.07	-4.0	0.07	-4.0	0.07	mA
F49,F50	IIh	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F51-F59	IIh	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F60	IIh	VII	X	-0.9	0.06	-0.9	0.06	-0.9	0.06	mA
F61-F65	IIh	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F66	IIh	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G1-G6	III	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G7-G22	III	VIII	X	-0.22	0.04	-0.22	0.04	-0.22	0.04	mA
G23-G25	III	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G26	III	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G27	III	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G28	III	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G29-G40	III	VIII	X	-0.5	0.02	-0.5	0.02	-0.5	0.02	mA
G41-G44	III	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G45	III	VIII	X	-2.7	0.09	-2.7	0.09	-2.7	0.09	mA
G46	III	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G47	III	VIII	X	-8.4	0.35	-8.4	0.35	-8.4	0.35	mA
G48	III	VIII	X	-4.0	0.07	-4.0	0.07	-4.0	0.07	mA
G49,G50	III	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G51-G59	III	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G60	III	VIII	X	-0.9	0.06	-0.9	0.06	-0.9	0.06	mA
G61-G65	III	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G66	III	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
H1-H36	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J36	Vol	X	X	0.0	0.5	0 0	0.5	0.0	0.5	V
K1-K16	Iozh	XI	X	-0.01	0.01	-0.01	0.01	-0.01	0.01	mA
K17,K18	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K19-K34	Iozl	XI	X	-0.01	0.01	-0.01	0.01	-0.01	0.01	mA
K35,K36	Iozl	XI	X	-0.05	0.05	-0 05	0.05	-0.05	0.05	mA
L1-L6	Vlc	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L7-L10	Vlc	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L11	Vlc	XII	S	-1.2	0.0	-1 2	0.0	-1.2	0.0	V
L12-L31	Vlc	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L32	Vlc	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L33-L50	Vlc	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
M1	PSFC	XIII	S	4.9	9.1	4 9	9.1	4.9	9.1	uF
M2	PSFC	XIII	S	7.0	13.0	7.0	13.0	7.0	13.0	uF
N	Icc	XIV	X	400	1150	400	1150	400	1150	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R1-R6	Cl/o	XVIII	Q	2	50	2	50	2	50	pF
R7-R11	Cl/o	XVIII	Q	2	75	2	75	2	75	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R12	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R13	C1/o	XVIII	Q	4	125	4	125	4	125	pF
R14	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R15,R16	C1/o	XVIII	Q	4	125	4	125	4	125	pF
R17	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R18-R20	C1/o	XVIII	Q	4	125	4	125	4	125	pF
R21	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R22-R24	C1/o	XVIII	Q	4	125	4	125	4	125	pF
R25-R28	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R29	C1/o	XVIII	Q	8	150	8	150	8	150	pF
R30	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R31	C1/o	XVIII	Q	2	175	2	175	2	175	pF
R32	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R33	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R34-R43	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R44	C1/o	XVIII	Q	6	100	6	100	6	100	pF
R45-R49	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R50	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R51-R54	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R55-R58	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R59-R65	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R66	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R67-R82	C1/o	XVIII	Q	4	75	4	75	4	75	pF
T1,T2	PSTR	XIX	Q	0	400	0	400	0	400	mV

- See footnotes at end of table.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/518-1 for class I and M28787/518-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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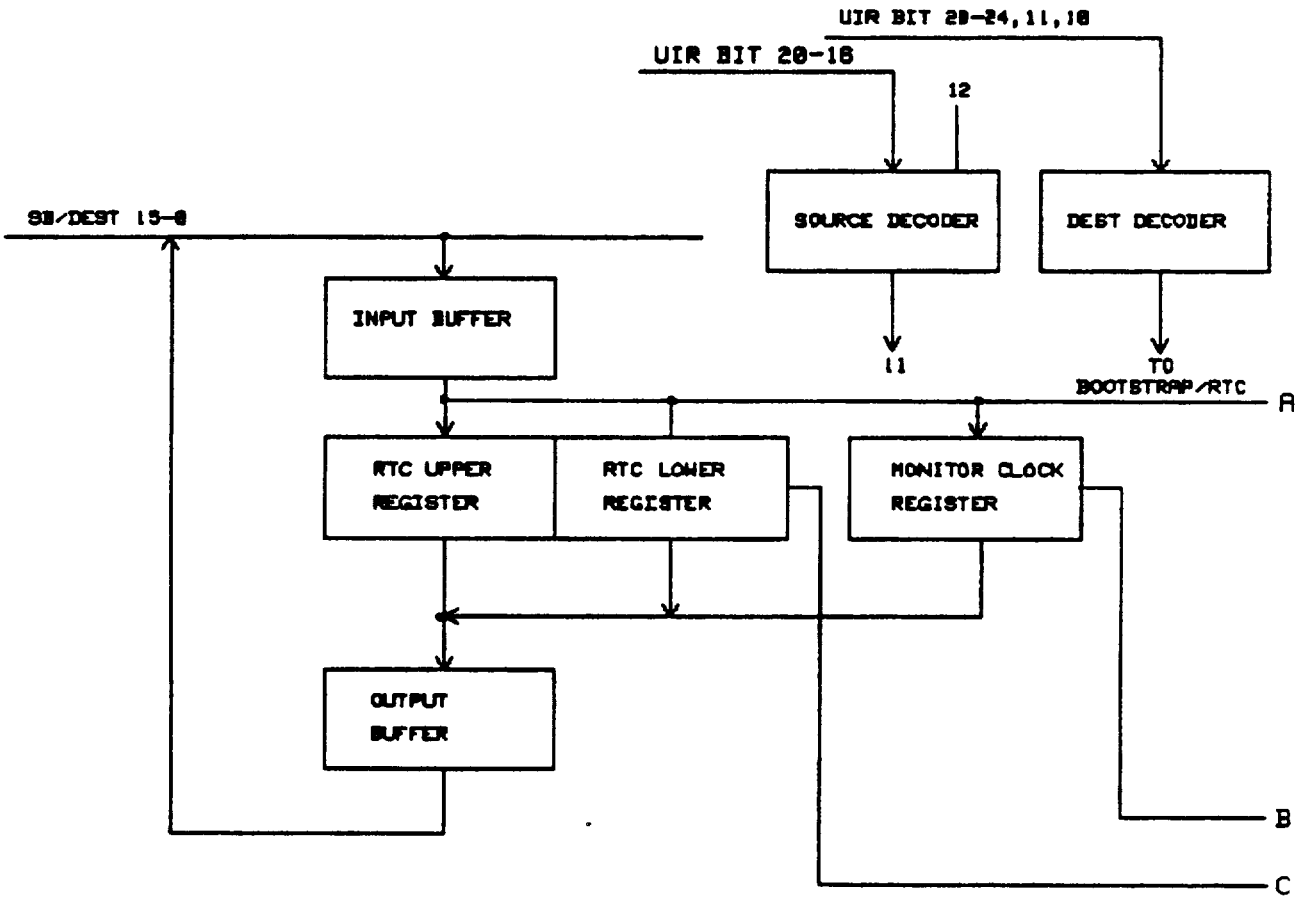


FIGURE 1. CWAF functional block diagram.

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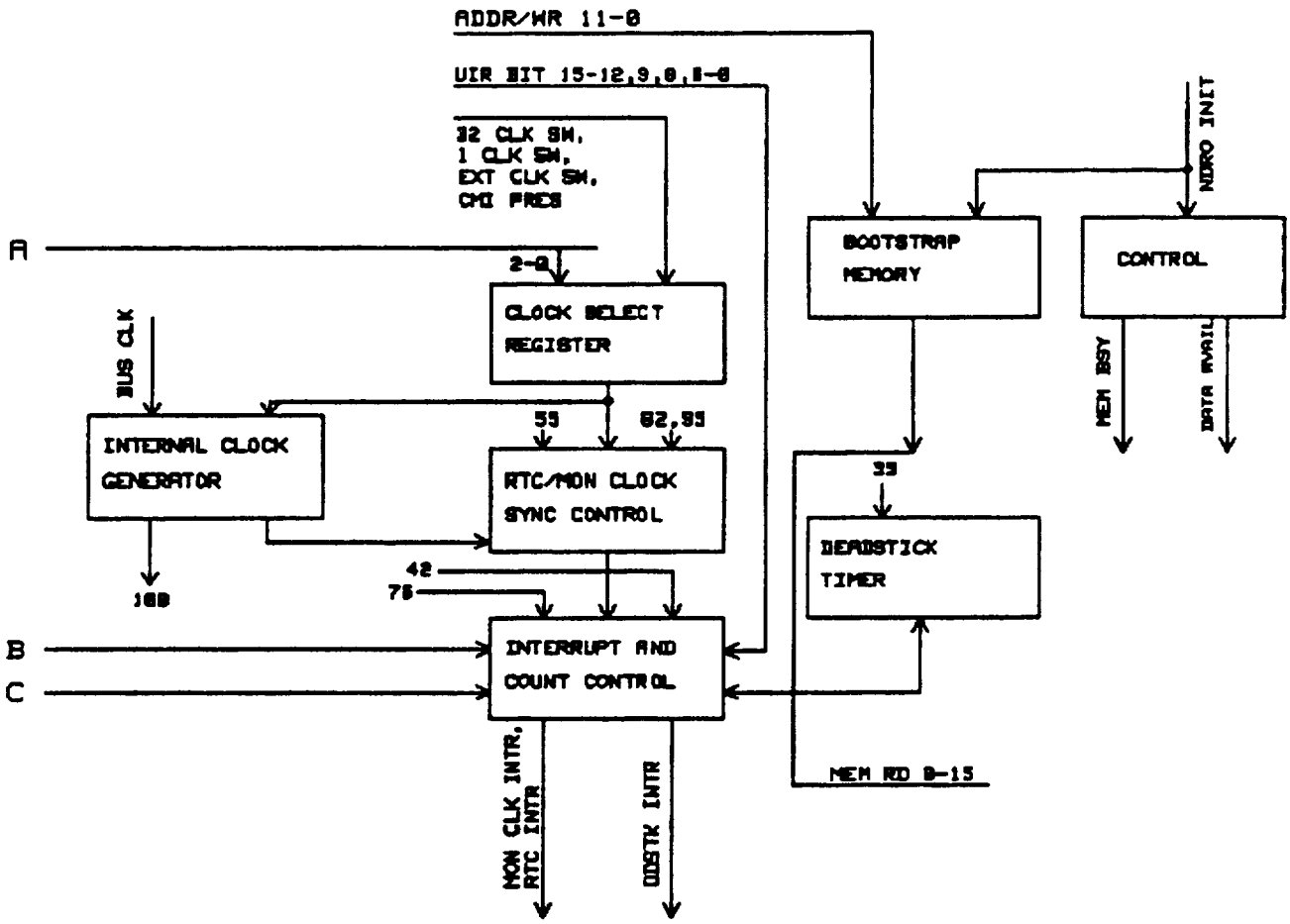


FIGURE 1. CWAF functional block diagram - Continued.

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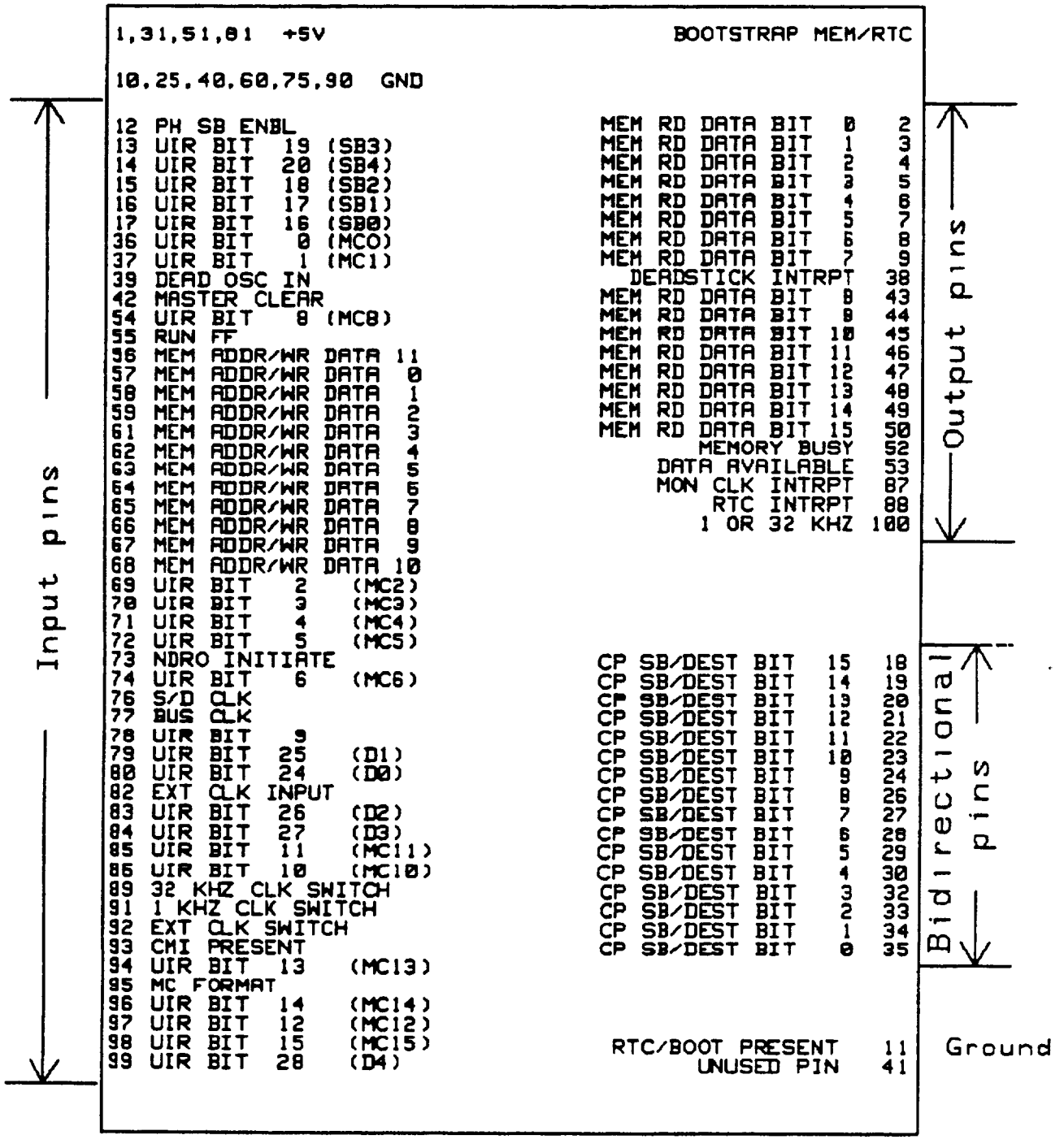


FIGURE 2. CWAF pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and will be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

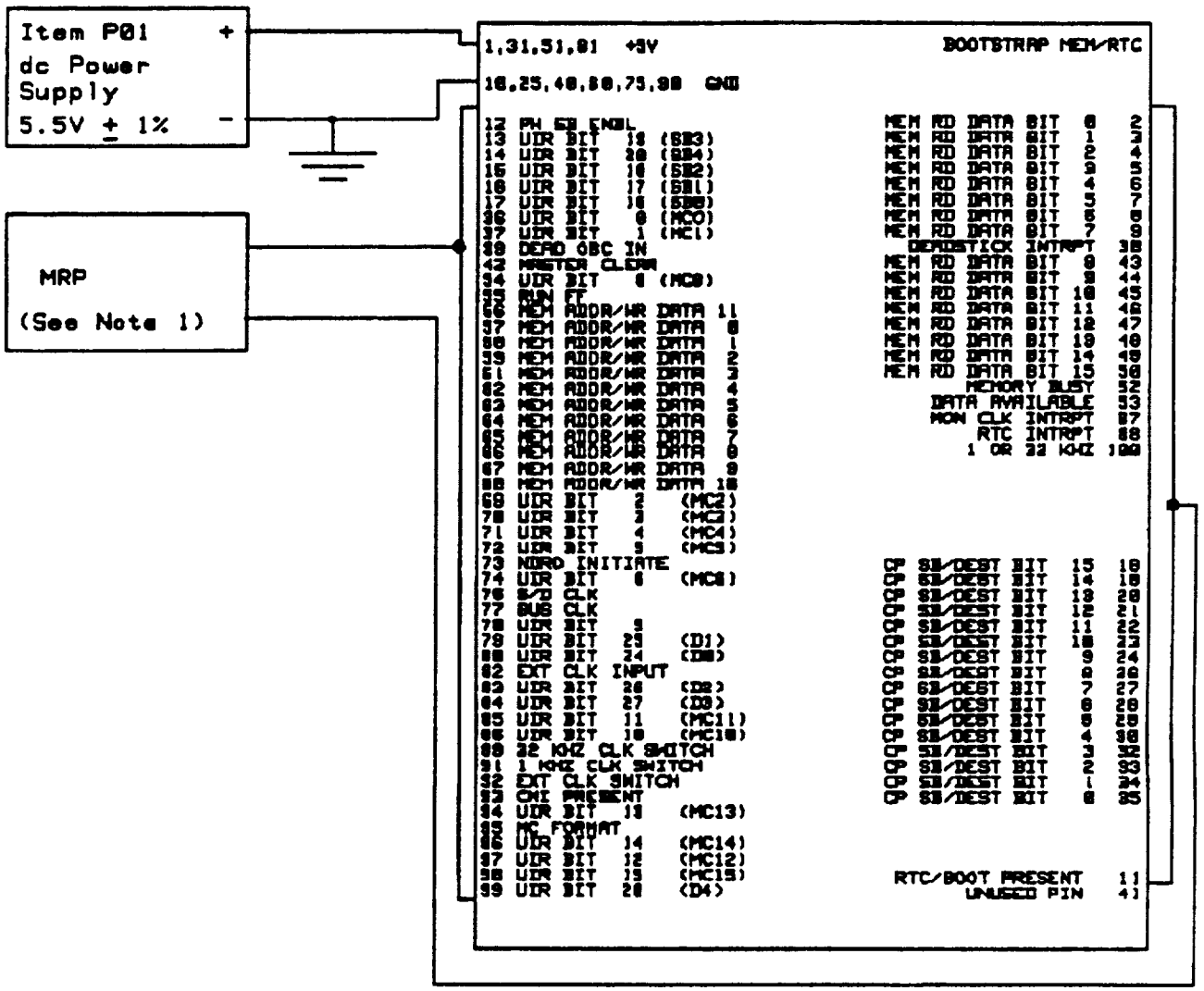
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. CWAF operating temperature and component temperature test circuit.

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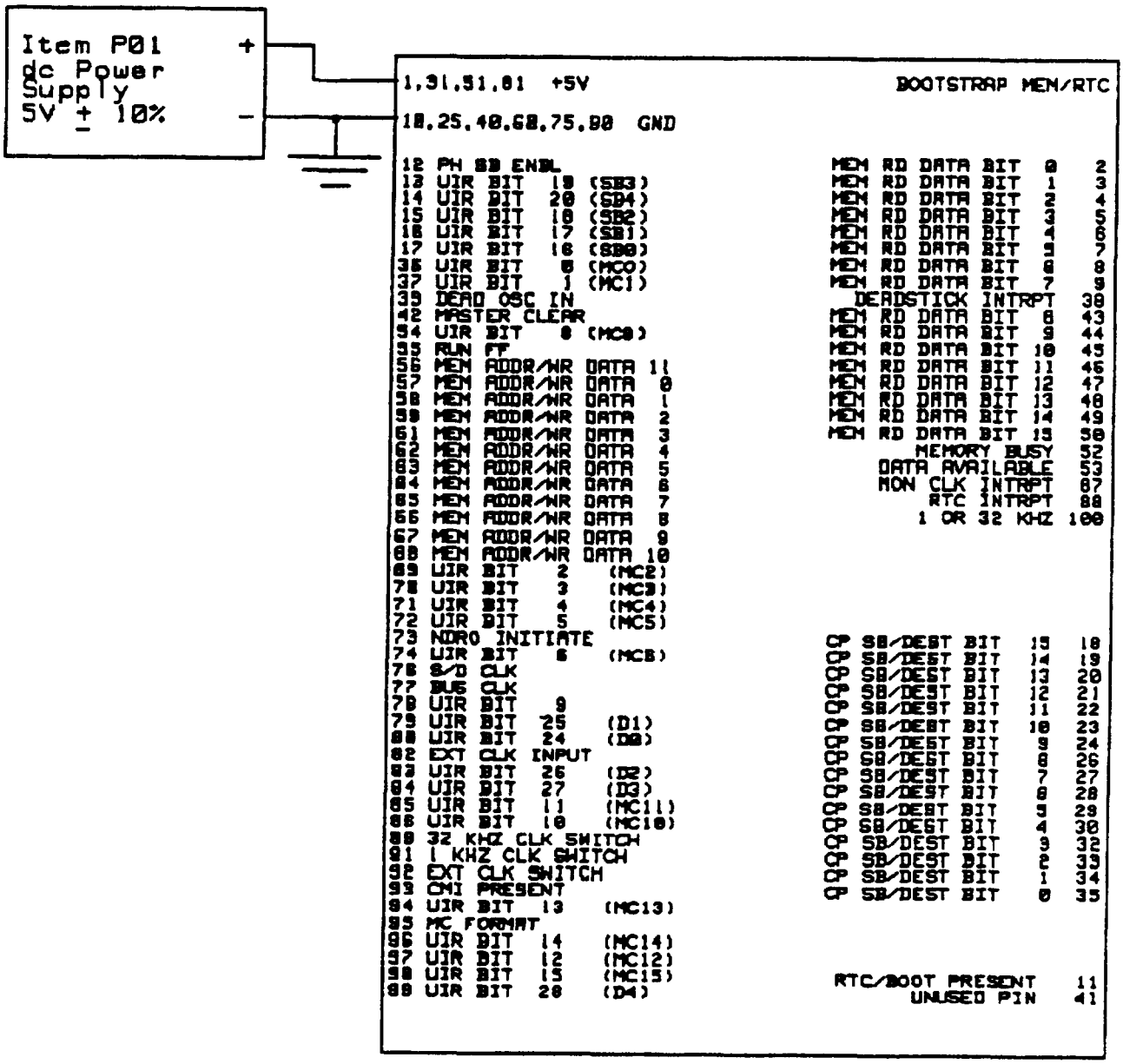


FIGURE 4. CWAF vibration test circuit.

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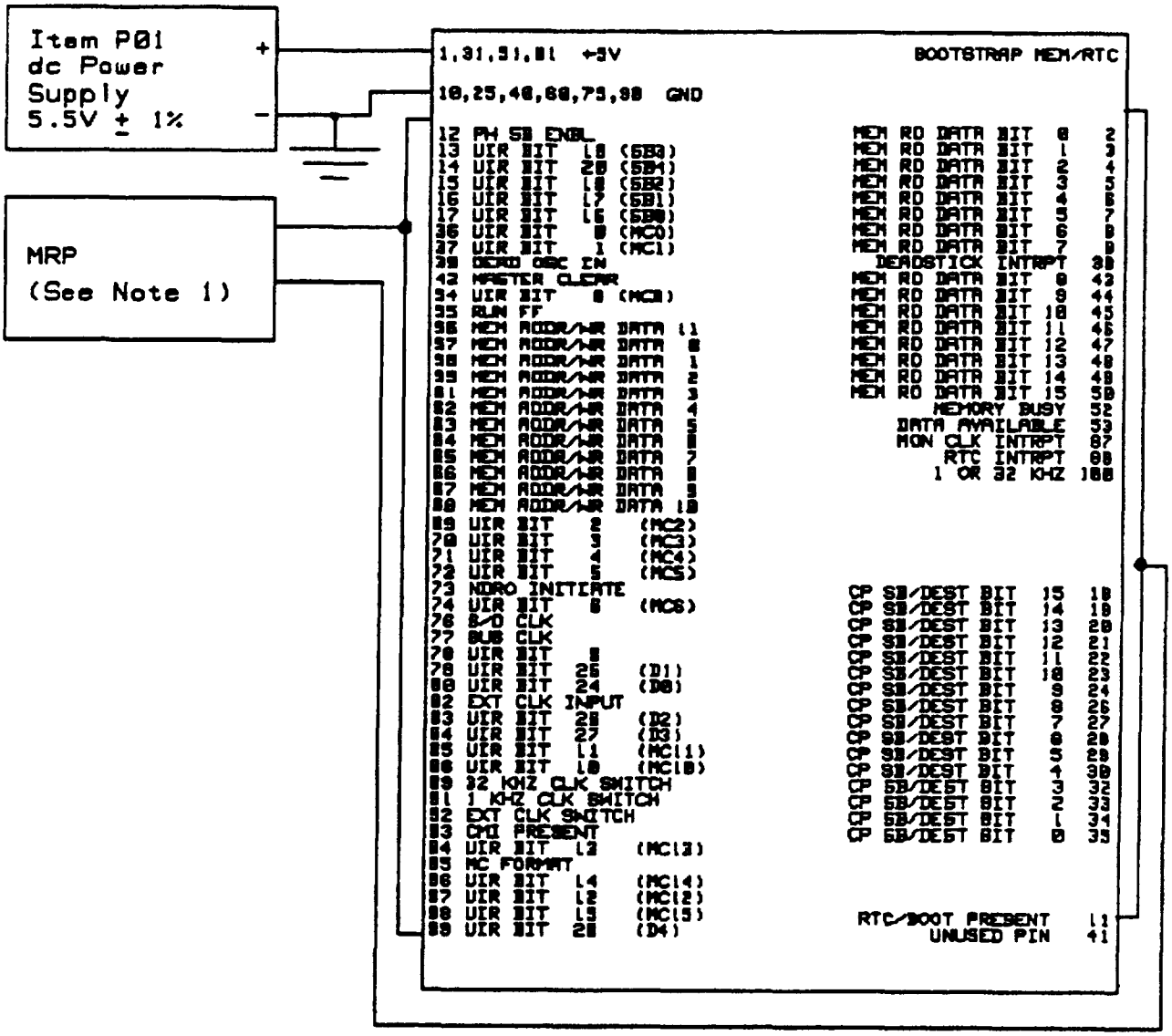
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. CWAF life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 6.325 W (max); 4.25 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
PH SB ENBL	-1.2	0.4	2.4	-0.6	0.02	0.1	50
UIR BIT 19	-1.2	0.4	2.4	-0.6	0.02	0.1	50
UIR BIT 20	-1.2	0.4	2.4	-0.6	0.02	0.1	50
UIR BIT 18	-1.2	0.4	2.4	-0.6	0.02	0.1	50
UIR BIT 17	-1.2	0.4	2.4	-0.6	0.02	0.1	50
UIR BIT 16	-1.2	0.4	2.4	-0.6	0.02	0.1	50
SB/DEST 15	NA	0.4	2.4	-0.22	0.04	NA	75
SB/DEST 14	NA	0.4	2.4	-0.22	0.04	NA	75
SB/DEST 13	NA	0.4	2.4	-0.22	0.04	NA	75
SB/DEST 12	NA	0.4	2.4	-0.22	0.04	NA	75
SB/DEST 11	NA	0.4	2.4	-0.22	0.04	NA	75
SB/DEST 10	NA	0.4	2.4	-0.22	0.04	NA	75
SB/DEST 9	NA	0.4	2.4	-0.22	0.04	NA	75
SB/DEST 8	NA	0.4	2.4	-0.22	0.04	NA	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
SB/DEST 7	NA	0.4	2.4	-0.22	0.04	NA	75
SB/DEST 6	NA	0.4	2.4	-0.22	0.04	NA	75
SB/DEST 5	NA	0.4	2.4	-0.22	0.04	NA	75
SB/DEST 4	NA	0.4	2.4	-0.22	0.04	NA	75
SB/DEST 3	NA	0.4	2.4	-0.22	0.04	NA	75
SB/DEST 2	NA	0.4	2.4	-0.22	0.04	NA	75
SB/DEST 1	NA	0.4	2.4	-0.22	0.04	NA	75
SB/DEST 0	NA	0.4	2.4	-0.22	0.04	NA	75
UIR BIT 0	-1.5	0.4	2.4	-0.4	0.02	0.1	75
UIR BIT 1	-1.5	0.4	2.4	-0.4	0.02	0.1	75
DEAD OSC IN	-1.5	0.4	2.4	-0.4	0.02	0.1	75
MSTR CLR	-1.5	0.4	2.4	-0.2	0.02	0.1	75
UIR BIT 8	-1.2	0.4	2.4	-2.0	0.05	1.0	75
RUN FF	-1.5	0.4	2.4	-0.2	0.02	0.1	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
ADDR/WR 11	-1.5	0.4	2.4	-0.5	0.02	2.0	125
ADDR/WR 0	-1.5	0.4	2.4	-0.5	0.02	2.0	100
ADDR/WR 1	-1.5	0.4	2.4	-0.5	0.02	2.0	125
ADDR/WR 2	-1.5	0.4	2.4	-0.5	0.02	2.0	125
ADDR/WR 3	-1.5	0.4	2.4	-0.5	0.02	2.0	100
ADDR/WR 4	-1.5	0.4	2.4	-0.5	0.02	2.0	125
ADDR/WR 5	-1.5	0.4	2.4	-0.5	0.02	2.0	125
ADDR/WR 6	-1.5	0.4	2.4	-0.5	0.02	2.0	125
ADDR/WR 7	-1.5	0.4	2.4	-0.5	0.02	2.0	100
ADDR/WR 8	-1.5	0.4	2.4	-0.5	0.02	2.0	125
ADDR/WR 9	-1.5	0.4	2.4	-0.5	0.02	2.0	125
ADDR/WR 10	-1.5	0.4	2.4	-0.5	0.02	2.0	125
UIR BIT 2	-1.5	0.4	2.4	-0.4	0.02	0.1	50
UIR BIT 3	-1.5	0.4	2.4	-0.4	0.02	0.1	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
UIR BIT 4	-1.5	0.4	2.4	-0.4	0.02	0.1	50
UIR BIT 5	-1.5	0.4	2.4	-0.4	0.02	0.1	50
NDRO INIT	-1.5	0.4	2.4	-2.7	0.09	3.1	150
UIR BIT 6	-1.5	0.4	2.4	-0.2	0.02	0.1	75
S/D CLK	-1.5	0.4	2.4	-8.4	0.35	3.4	175
BUS CLK	-1.2	0.4	2.4	-4.0	0.07	2.0	75
UIR BIT 9	-1.5	0.4	2.4	-0.2	0.02	0.1	75
UIR BIT 25	-1.5	0.4	2.4	-0.2	0.02	0.1	50
UIR BIT 24	-1.5	0.4	2.4	-0.4	0.02	0.1	50
EXT CLK IN	-1.5	0.4	2.4	-0.4	0.02	0.1	50
UIR BIT 26	-1.5	0.4	2.4	-0.4	0.02	0.1	50
UIR BIT 27	-1.5	0.4	2.4	-0.4	0.02	0.1	50
UIR BIT 11	-1.5	0.4	2.4	-0.4	0.02	0.1	50
UIR BIT 10	-1.5	0.4	2.4	-0.4	0.02	0.1	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
32 CLK SW	-1.5	0.4	2.4	-0.4	0.02	0.1	50
1 CLK SW	-1.5	0.4	2.4	-0.4	0.02	0.1	50
EXT CLK SW	-1.5	0.4	2.4	-0.4	0.02	0.1	50
CMI PRES	-1.5	0.4	2.4	-0.8	0.06	0.3	100
UIR BIT 13	-1.5	0.4	2.4	-0.4	0.02	0.1	50
MC FMT	-1.5	0.4	2.4	-0.4	0.02	0.1	50
UIR BIT 14	-1.5	0.4	2.4	-0.4	0.02	0.1	50
UIR BIT 12	-1.5	0.4	2.4	-0.4	0.02	0.1	50
UIR BIT 15	-1.5	0.4	2.4	-0.4	0.02	0.1	50
UIR BIT 28	-1.5	0.4	2.4	-0.2	0.02	0.1	75

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
MEM RD 0	2.4	0.5	-0.4	6.0	0.01	0.01	-12.0	50
MEM RD 1	2.4	0.5	-0.4	6.0	0.01	0.01	-12.0	50
MEM RD 2	2.4	0.5	-0.4	6.0	0.01	0.01	-12.0	50
MEM RD 3	2.4	0.5	-0.4	6.0	0.01	0.01	-12.0	50
MEM RD 4	2.4	0.5	-0.4	6.0	0.01	0.01	-12.0	75
MEM RD 5	2.4	0.5	-0.4	6.0	0.01	0.01	-12.0	75
MEM RD 6	2.4	0.5	-0.4	6.0	0.01	0.01	-12.0	75
MEM RD 7	2.4	0.5	-0.4	6.0	0.01	0.01	-12.0	75
SB/DEST 15	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
SB/DEST 14	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
SB/DEST 13	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
SB/DEST 12	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
SB/DEST 11	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
SB/DEST 10	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
SB/DEST 9	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
SB/DEST 8	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
SB/DEST 7	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
SB/DEST 6	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
SB/DEST 5	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
SB/DEST 4	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
SB/DEST 3	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
SB/DEST 2	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
SB/DEST 1	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
SB/DEST 0	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
DDSTK INTR	2.4	0.5	-0.16	1.6	NA	NA	-15.0	NA
MEM RD 8	2.4	0.5	-0.4	6.0	0.01	0.01	-12.0	50
MEM RD 9	2.4	0.5	-0.4	6.0	0.01	0.01	-12.0	50
MEM RD 10	2.4	0.5	-0.4	6.0	0.01	0.01	-12.0	50

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
MEM RD 11	2.4	0.5	-0.4	6.0	0.01	0.01	-12.0	50
MEM RD 12	2.4	0.5	-0.4	6.0	0.01	0.01	-12.0	50
MEM RD 13	2.4	0.5	-0.4	6.0	0.01	0.01	-12.0	50
MEM RD 14	2.4	0.5	-0.4	6.0	0.01	0.01	-12.0	50
MEM RD 15	2.4	0.5	-0.4	6.0	0.01	0.01	-12.0	75
MEM BSY	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
DATA AVAIL	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
MON CLK INTR	2.4	0.5	-0.16	1.6	NA	NA	-15.0	NA
RTC INTR	2.4	0.5	-0.16	1.6	NA	NA	-15.0	NA
1 OR 32 KHZ	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tsu	ADDR/WR 11 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 0 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 1 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 2 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 3 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 4 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 5 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 6 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 7 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 8 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 9 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 10 to NDRO INIT	5	5	5	-	-	-	nS	12
2	tpd	BUS CLK to MEM BSY	-	-	-	38	38	38	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	NDRO INIT to MEM RD 0	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 1	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 2	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 3	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 4	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 5	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 6	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 7	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 8	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 9	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 10	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 11	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 12	-	-	-	109	109	109	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	NDRO INIT to MEM RD 13	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 14	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 15	-	-	-	109	109	109	nS	12
4	tpd	UIR BIT 19 to SB/DEST 15	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 14	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 13	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 12	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 11	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 10	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 9	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 8	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 7	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 6	-	-	-	59	59	59	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	UIR BIT 19 to SB/DEST 5	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 4	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 3	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 2	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 1	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 0	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 15	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 14	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 13	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 12	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 11	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 10	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 9	-	-	-	59	59	59	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	UIR BIT 20 to SB/DEST 8	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 7	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 6	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 5	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 4	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 3	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 2	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 1	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 0	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 15	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 14	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 13	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 12	-	-	-	59	59	59	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	UIR BIT 18 to SB/DEST 11	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 10	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 9	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 8	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 7	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 6	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 5	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 4	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 3	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 2	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 1	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 0	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 15	-	-	-	59	59	59	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	UIR BIT 17 to SB/DEST 14	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 13	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 12	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 11	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 10	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 9	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 8	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 7	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 6	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 5	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 4	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 3	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 2	-	-	-	59	59	59	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	UIR BIT 17 to SB/DEST 1	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 0	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 15	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 14	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 13	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 12	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 11	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 10	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 9	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 8	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 7	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 6	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 5	-	-	-	59	59	59	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	UIR BIT 16 to SB/DEST 4	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 3	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 2	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 1	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 0	-	-	-	59	59	59	nS	12
5	tpd	S/D CLK to MON CLK INTR	-	-	-	53	53	53	nS	12
5	tpd	S/D CLK to RTC INTR	-	-	-	53	53	53	nS	12
6	tpd	DEAD OSC IN to DDSTK INTR	-	-	-	80	80	80	nS	12
7	tpd	BUS CLK to DATA AVAIL	-	-	-	52	52	52	nS	12

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6.4 Module function. The Real Time Clock (RTC) and monitor clock features provide two program controlled interrupts via two registers, one 32-bit register (divided into two 16-bit registers designated RTC upper and RTC lower) used as RTC count-up storage and the other a 16-bit register as monitor clock count-down storage. An RTC oscillator, which has an accuracy of ± 2 counts in test seconds, runs continuously and controls the counting speed of both registers. Selectable rates for the RTC are 1 KHz or 32 KHz. An external clock oscillator with a frequency in the 0 to 50 KHz range may be used instead of the internal oscillator. The registers are loaded and their incrementing and decrementing are enabled and disabled under program control by means of the unary control macroinstruction (Op Code 03 base 8). The RTC overflow interrupt is generated when RTC lower changes from all ones to all zeros. RTC upper effectively keeps track of how many times this overflow has occurred. The monitor clock interrupt is generated when the monitor count reaches zero.

6.4.1 RTC register. The 32-bit RTC register is divided into two 16-bit registers: RTC upper and RTC lower. The register operates in conjunction with an internal generated clock (1 KHz or 32 KHz) or an external clock (0 to 50 KHz). The RTC, when enabled, increments by one on each clock pulse. When RTC lower changes from all ones to all zeros (in other words, reaches its maximum count), a RTC overflow interrupt is generated (if enabled). RTC upper effectively keeps track of how many times this overflow has occurred. The register is loaded or read via the source B/Destination bus. RTC upper is loaded by a microinstruction with D = 0C and bits 11 and 10 = 2. RTC lower is loaded by a microinstruction with D = 0C and bits 11 and 10 = 1. RTC upper is read by a microinstruction with SB = 17 and RTC lower with SB = 16.

6.4.2 Monitor clock register. The 16 bit monitor clock register is mainly used for generating time dependent interrupts. When enabled, it decrements at the same frequency selected for the RTC register. When the monitor count reaches zero, a monitor clock interrupt is generated (if enabled). The register is loaded or read via the source B/Destination bus. The register is loaded by a microinstruction with D = 0C and bits 11 and 10 = 0. The register is read by a microinstruction with SB = 15.

6.4.3 Source decoder. The source decoder translates the microinstruction Source B (SB) field (bits 20-16) to provide

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output enables to the RTC upper and lower registers and monitor clock register. The decoder is enabled by the Phase Source B enable.

6.4.4 Destination decoder. The destination decoder translates the microinstruction D field (bits 28-24) and bits 11 and 10 to provide load enables to the RTC upper and lower registers, monitor clock register, and clock select register.

6.4.5 Clock select register. The clock select register holds the clock operating source (1 KHz clock, 32 KHz clock, or external clock) information. The source for the selection is determined by the presence of the Control and Maintenance Interface (CMI). If the CMI is present, the source is source B/Destination bus bits 2-0 and the register is loaded by a microinstruction with D = 0C and bits 11 and 10 = 3. Bits 2-0 select the clock as follows (binary):

011 - 32 KHz
101 - 1 KHz
110 - External Clock

If CMI is not present, the register is loaded from an external source (usually switches) via input pins 89, 91, and 92.

6.4.6 Internal clock generator. The internal clock generator consists of a 16-bit counter that generates a 1 KHz or 32 KHz (selected by clock select register) output pulse. The pulses are derived from the bus clock (20 MHz).

6.4.7 RTC/monitor clock sync control. Sync control provides a 50 nanosecond pulse to advance the RTC or decrement the monitor clock on the positive going edge of the selected clock (1 KHz, 32 KHz, or external). The pulse is synchronized with the negative going edge of the Source/Destination (S/D) clock.

6.4.8 Deadstick timer. The deadstick timer consists of a 4-bit counter. The timer is reset by a format 3 microinstruction with MC = E and bit 8 = 1. If the timer is not reset within one second, a deadstick interrupt is generated. The timer is disabled by a master clear until reset by the microinstruction.

6.4.9 Interrupt and count control. The interrupt and count control circuits enable and disable the RTC and monitor clocks and interrupts under microinstruction control. A format 3

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microinstruction with MC = D or E along with selected Microinstruction Bits 9-0 (MCU) set or clear individual flip-flops to provide the enables or disables. Refer to the microinstruction repertoire for MCU field interpretation. The monitor clock and RTC interrupts are used to generate class II macro interrupts. The deadstick interrupt is used to generate a maintenance processor interrupt.

6.4.10 Description of I/O pins.

Signal name	Pin number	I/O	Description
MEM RD 0	2	O	MEM RD DATA BIT 0
MEM RD 1	3	O	MEM RD DATA BIT 1
MEM RD 2	4	O	MEM RD DATA BIT 2
MEM RD 3	5	O	MEM RD DATA BIT 3
MEM RD 4	6	O	MEM RD DATA BIT 4
MEM RD 5	7	O	MEM RD DATA BIT 5
MEM RD 6	8	O	MEM RD DATA BIT 6
MEM RD 7	9	O	MEM RD DATA BIT 7
PH SB ENBL	12	I	PH SB ENBL
UIR BIT 19	13	I	UIR BIT 19 (SB3)
UIR BIT 20	14	I	UIR BIT 20 (SB4)
UIR BIT 18	15	I	UIR BIT 18 (SB2)
UIR BIT 17	16	I	UIR BIT 17 (SB1)
UIR BIT 16	17	I	UIR BIT 16 (SB0)
SB/DEST 15	18	I/O	CP SB/DEST BIT 15

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6.4.10 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
SB/DEST 14	19	I/O	CP SB/DEST BIT 14
SB/DEST 13	20	I/O	CP SB/DEST BIT 13
SB/DEST 12	21	I/O	CP SB/DEST BIT 12
SB/DEST 11	22	I/O	CP SB/DEST BIT 11
SB/DEST 10	23	I/O	CP SB/DEST BIT 10
SB/DEST 9	24	I/O	CP SB/DEST BIT 9
SB/DEST 8	26	I/O	CP SB/DEST BIT 8
SB/DEST 7	27	I/O	CP SB/DEST BIT 7
SB/DEST 6	28	I/O	CP SB/DEST BIT 6
SB/DEST 5	29	I/O	CP SB/DEST BIT 5
SB/DEST 4	30	I/O	CP SB/DEST BIT 4
SB/DEST 3	32	I/O	CP SB/DEST BIT 3
SB/DEST 2	33	I/O	CP SB/DEST BIT 2
SB/DEST 1	34	I/O	CP SB/DEST BIT 1
SB/DEST 0	35	I/O	CP SB/DEST BIT 0
UIR BIT 0	36	I	UIR BIT 0 (MC0)
UIR BIT 1	37	I	UIR BIT 1 (MC1)
DDSTK INTR	38	O	DEADSTICK INTRPT
DEAD OSC IN	39	I	DEAD OSC IN
MSTR CLR	42	I	MASTER CLEAR

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6.4.10 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MEM RD 8	43	O	MEM RD DATA BIT 8
MEM RD 9	44	O	MEM RD DATA BIT 9
MEM RD 10	45	O	MEM RD DATA BIT 10
MEM RD 11	46	O	MEM RD DATA BIT 11
MEM RD 12	47	O	MEM RD DATA BIT 12
MEM RD 13	48	O	MEM RD DATA BIT 13
MEM RD 14	49	O	MEM RD DATA BIT 14
MEM RD 15	50	O	MEM RD DATA BIT 15
MEM BSY	52	O	MEMORY BUSY
DATA AVAIL	53	O	DATA AVAILABLE
UIR BIT 8	54	I	UIR BIT 8 (MC8)
RUN FF	55	I	RUN FF
ADDR/WR 11	56	I	MEM ADDR/WR DATA 11
ADDR/WR 0	57	I	MEM ADDR/WR DATA 0
ADDR/WR 1	58	I	MEM ADDR/WR DATA 1
ADDR/WR 2	59	I	MEM ADDR/WR DATA 2
ADDR/WR 3	61	I	MEM ADDR/WR DATA 3
ADDR/WR 4	62	I	MEM ADDR/WR DATA 4
ADDR/WR 5	63	I	MEM ADDR/WR DATA 5
ADDR/WR 6	64	I	MEM ADDR/WR DATA 6

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6.4.10 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
ADDR/WR 7	65	I	MEM ADDR/WR DATA 7
ADDR/WR 8	66	I	MEM ADDR/WR DATA 8
ADDR/WR 9	67	I	MEM ADDR/WR DATA 9
ADDR/WR 10	68	I	MEM ADDR/WR DATA 10
UIR BIT 2	69	I	UIR BIT 2 (MC2)
UIR BIT 3	70	I	UIR BIT 3 (MC3)
UIR BIT 4	71	I	UIR BIT 4 (MC4)
UIR BIT 5	72	I	UIR BIT 5 (MC5)
NDRO INIT	73	I	NDRO INITIATE
UIR BIT 6	74	I	UIR BIT 6 (MC6)
S/D CLK	76	I	SOURCE/DESTINATION CLK
BUS CLK	77	I	BUS CLK
UIR BIT 9	78	I	UIR BIT 9
UIR BIT 25	79	I	UIR BIT 25 (D1)
UIR BIT 24	80	I	UIR BIT 24 (D0)
EXT CLK IN	82	I	EXT CLK INPUT
UIR BIT 26	83	I	UIR BIT 26 (D2)
UIR BIT 27	84	I	UIR BIT 27 (D3)
UIR BIT 11	85	I	UIR BIT 11 (MC11)
UIR BIT 10	86	I	UIR BIT 10 (MC10)

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6.4.10 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MON CLK INTR	87	O	MON CLK INTRPT
RTC INTR	88	O	RTC INTRPT
32 CLK SW	89	I	32 KHz CLK SWITCH
1 CLK SW	91	I	1 KHz CLK SWITCH
EXT CLK SW	92	I	EXT CLK SWITCH
CMI PRES	93	I	CMI PRESENT
UIR BIT 13	94	I	UIR BIT 13 (MC13)
MC FMT	95	I	MC FORMAT
UIR BIT 14	96	I	UIR BIT 14 (MC14)
UIR BIT 12	97	I	UIR BIT 12 (MC12)
UIR BIT 15	98	I	UIR BIT 15 (MC15)
UIR BIT 28	99	I	UIR BIT 28 (D4)
1 OR 32 KHZ	100	O	1 OR 32 KHZ
RTC/BOOT PRESENT	11	GND	RTC/BOOT PRESENT

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N714)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to	Connect B to	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C,41	Frame	10	INF	MOhm	1-8
A2	Isolation, used to unused pins	C	41	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1,A2
Isolation measurement	NA	± 500 KOhm	A1,A2

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

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NOTES: - Continued.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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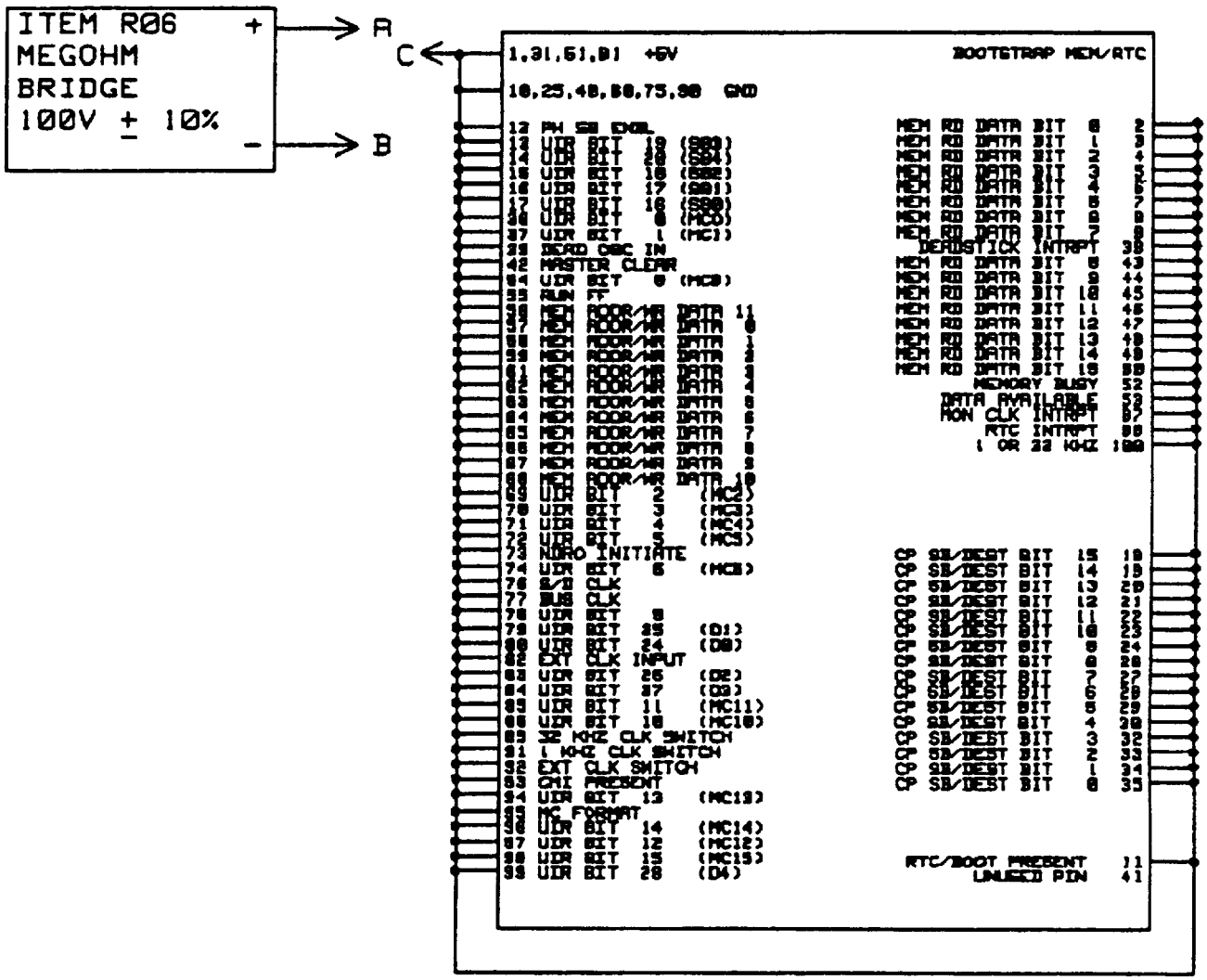


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to	Connect B to	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, ground	10	25	0	1	Ohm	1-5
B2	Continuity, ground	10	40				
B3	Continuity, ground	60	75				
B4	Continuity, ground	60	90				
B5	Continuity, +5V	1	31				
B6	Continuity, +5V	51	81				
B7	Continuity, RTC/BOOT PRESENT to ground	10	11	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B7

2. The ohmmeter used shall be item R07.

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NOTES: - Continued.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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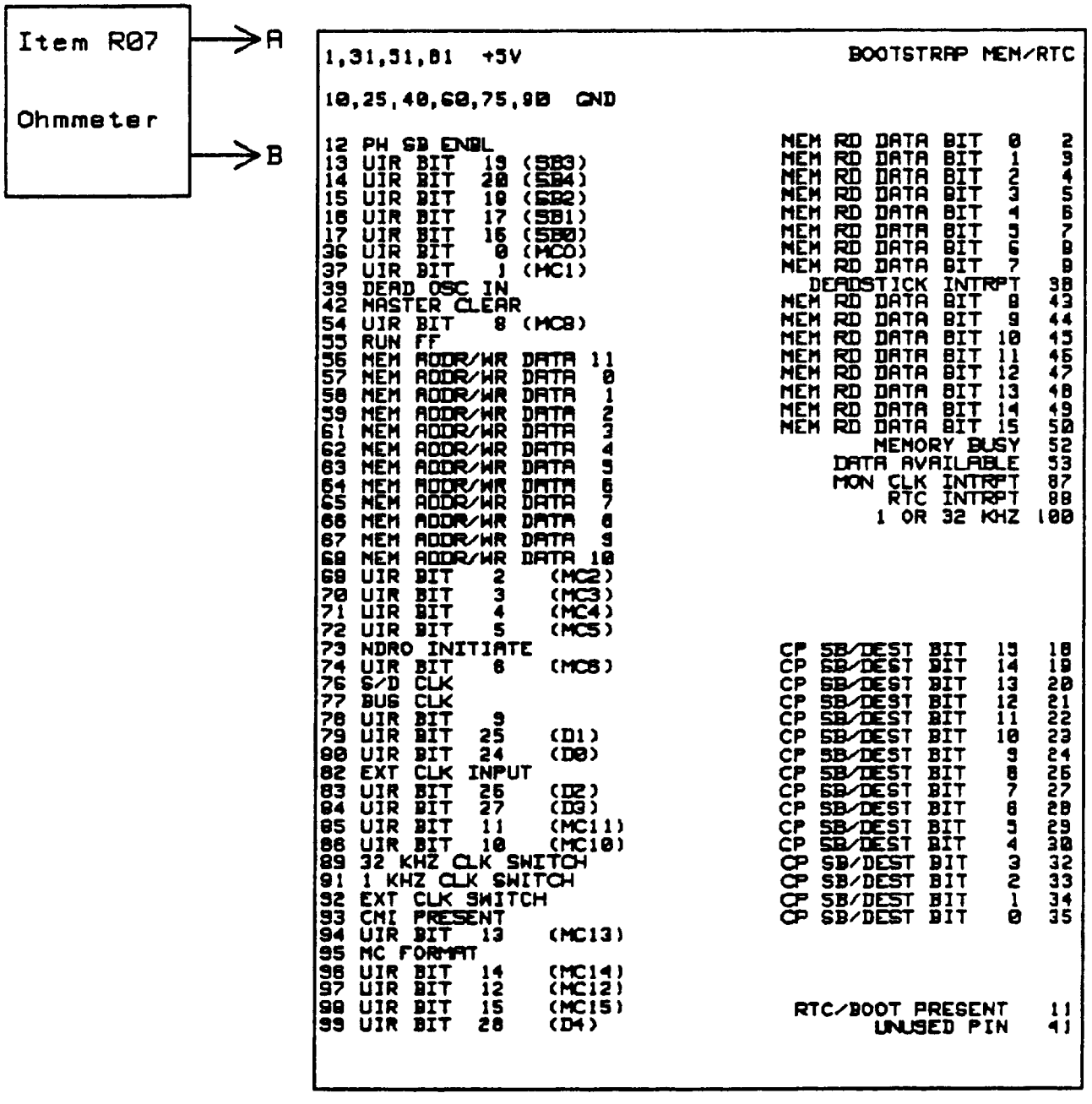


FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	2.07 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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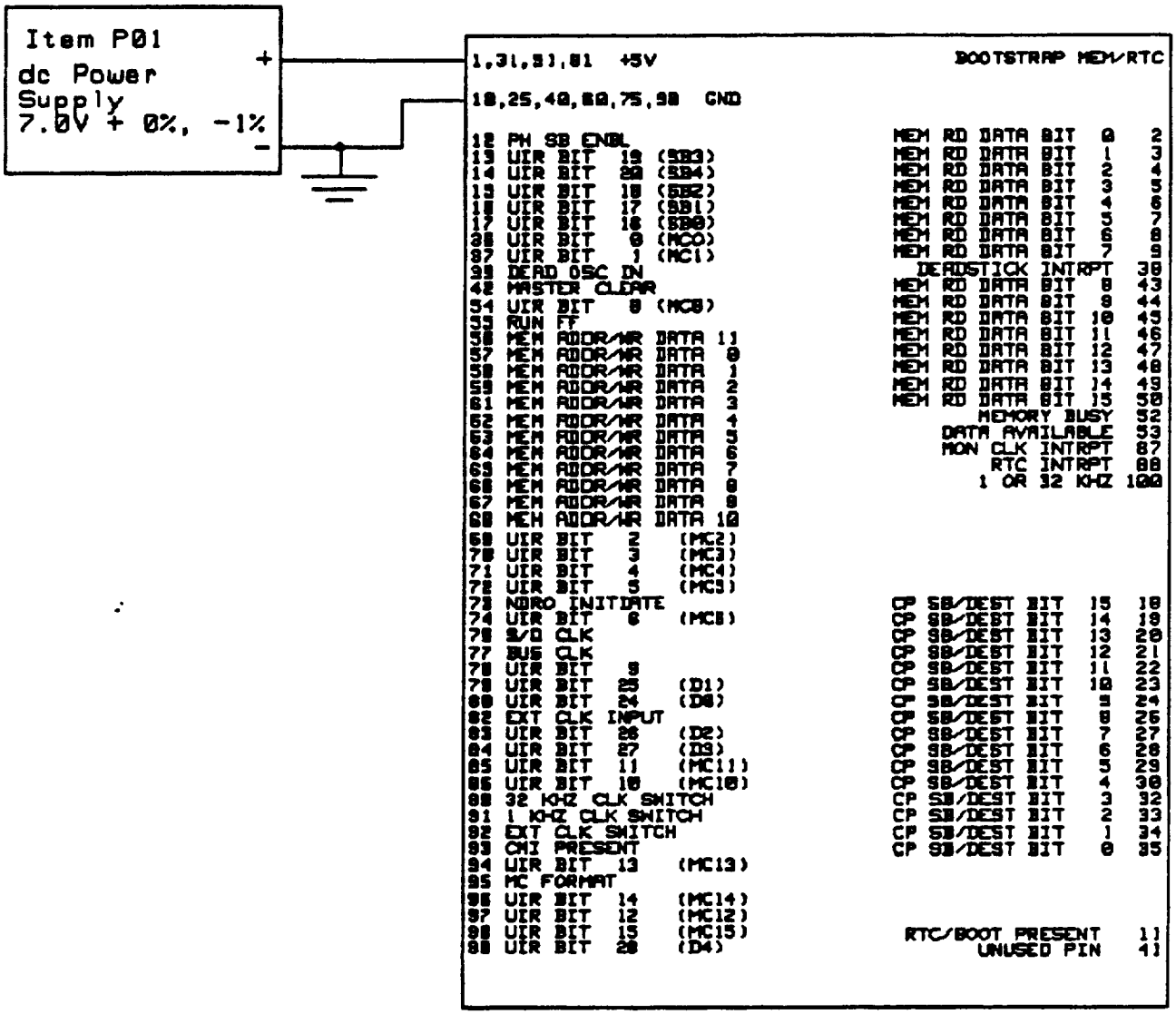


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	PH SB ENBL	12	5.5	0.0	0.1	mA	1-3
D2	UIR BIT 19	13					
D3	UIR BIT 20	14					
D4	UIR BIT 18	15					
D5	UIR BIT 17	16					
D6	UIR BIT 16	17					
D7	UIR BIT 0	36					
D8	UIR BIT 1	37					
D9	DEAD OSC IN	39					
D10	MSTR CLR	42	5.5	0.0	0.1	mA	1-3
D11	UIR BIT 8	54	5.5	0.0	1.0	mA	1-3
D12	RUN FF	55	5.5	0.0	0.1	mA	1-3
D13	ADDR/WR 11	56	5.5	0.0	2.0	mA	1-3
D14	ADDR/WR 0	57					
D15	ADDR/WR 1	58					
D16	ADDR/WR 2	59					
D17	ADDR/WR 3	61					
D18	ADDR/WR 4	62					
D19	ADDR/WR 5	63					
D20	ADDR/WR 6	64					
D21	ADDR/WR 7	65	5.5	0.0	2.0	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D22	ADDR/WR 8	66	5.5	0.0	2.0	mA	1-3
D23	ADDR/WR 9	67					
D24	ADDR/WR 10	68	5.5	0.0	2.0	mA	1-3
D25	UIR BIT 2	69	5.5	0.0	0.1	mA	1-3
D26	UIR BIT 3	70					
D27	UIR BIT 4	71					
D28	UIR BIT 5	72	5.5	0.0	0.1	mA	1-3
D29	NDRO INIT	73	5.5	0.0	3.1	mA	1-3
D30	UIR BIT 6	74	5.5	0.0	0.1	mA	1-3
D31	S/D CLK	76	5.5	0.0	3.4	mA	1-3
D32	BUS CLK	77	5.5	0.0	2.0	mA	1-3
D33	UIR BIT 9	78	5.5	0.0	0.1	mA	1-3
D34	UIR BIT 25	79					
D35	UIR BIT 24	80					
D36	EXT CLK IN	82					
D37	UIR BIT 26	83					
D38	UIR BIT 27	84					
D39	UIR BIT 11	85					
D40	UIR BIT 10	86					
D41	32 CLK SW	89					
D42	1 CLK SW	91	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D43	EXT CLK SW	92	5.5	0.0	0.1	mA	1-3
D44	CMI PRES	93	5.5	0.0	0.3	mA	1-3
D45	UIR BIT 13	94	5.5	0.0	0.1	mA	1-3
D46	MC FMT	95					
D47	UIR BIT 14	96					
D48	UIR BIT 12	97					
D49	UIR BIT 15	98					
D50	UIR BIT 28	99	5.5	0.0	0.1	mA	1-3

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D50
Power supply current limit	1.73 A	NA	D1-D50
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D50

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	D1-D50
low, (VL)	0.0 V	± 100 mV	D1-D50
Input current measurement	NA	± 5 uA	D1-D10
	NA	± 50 uA	D11
	NA	± 5 uA	D12
	NA	± 100 uA	D13-D24
	NA	± 5 uA	D25-D28
	NA	± 155 uA	D29
	NA	± 5 uA	D30
	NA	± 170 uA	D31
	NA	± 100 uA	D32
	NA	± 5 uA	D33-D43
	NA	± 15 uA	D44
	NA	± 5 uA	D45-D50

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	MEM RD 0	2	0.0	-85.0	-12.0	mA	1-4
E2	MEM RD 1	3					
E3	MEM RD 2	4					
E4	MEM RD 3	5					
E5	MEM RD 4	6					
E6	MEM RD 5	7					
E7	MEM RD 6	8					
E8	MEM RD 7	9	0.0	-85.0	-12.0	mA	1-4
E9	SB/DEST 15	18	0.0	-220.0	-30.0	mA	1-4
E10	SB/DEST 14	19					
E11	SB/DEST 13	20					
E12	SB/DEST 12	21					
E13	SB/DEST 11	22					
E14	SB/DEST 10	23					
E15	SB/DEST 9	24					
E16	SB/DEST 8	26					
E17	SB/DEST 7	27					
E18	SB/DEST 6	28					
E19	SB/DEST 5	29					
E20	SB/DEST 4	30					
E21	SB/DEST 3	32	0.0	-220.0	-30.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E22	SB/DEST 2	33	0.0	-220.0	-30.0	mA	1-4
E23	SB/DEST 1	34					
E24	SB/DEST 0	35	0.0	-220.0	-30.0	mA	1-4
E25	DDSTK INTR	38	0.0	-100.0	-15.0	mA	1-4
E26	MEM RD 8	43	0.0	-85.0	-12.0	mA	1-4
E27	MEM RD 9	44					
E28	MEM RD 10	45					
E29	MEM RD 11	46					
E30	MEM RD 12	47					
E31	MEM RD 13	48					
E32	MEM RD 14	49					
E33	MEM RD 15	50	0.0	-85.0	-12.0	mA	1-4
E34	MEM BSY	52	0.0	-325.0	-100.0	mA	1-4
E35	DATA AVAIL	53	0.0	-325.0	-100.0	mA	1-4
E36	MON CLK INTR	87	0.0	-100.0	-15.0	mA	1-4
E37	RTC INTR	88	0.0	-100.0	-15.0	mA	1-4
E38	1 OR 32 KHZ	100	0.0	-220.0	-30.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E38
Power supply current limit	1.73 A	NA	E1-E38
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E38
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	E1-E38
low, (VL)	0.0 V	± 100 mV	E1-E38
Output current measurement	NA	± 3.65 mA	E1-E8
	NA	± 9.50 mA	E9-E24
	NA	± 4.25 mA	E25
	NA	± 3.65 mA	E26-E33
	NA	± 11.2 mA	E34,E35
	NA	± 4.25 mA	E36,E37
	NA	± 9.50 mA	E38

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	PH SB ENBL	12	2.4	-0.6	0.02	mA	1-3
F2	UIR BIT 19	13					
F3	UIR BIT 20	14					
F4	UIR BIT 18	15					
F5	UIR BIT 17	16					
F6	UIR BIT 16	17	2.4	-0.6	0.02	mA	1-3
F7	SB/DEST 15	18	2.4	-0.22	0.04	mA	1-3
F8	SB/DEST 14	19					
F9	SB/DEST 13	20					
F10	SB/DEST 12	21					
F11	SB/DEST 11	22					
F12	SB/DEST 10	23					
F13	SB/DEST 9	24					
F14	SB/DEST 8	26					
F15	SB/DEST 7	27					
F16	SB/DEST 6	28					
F17	SB/DEST 5	29					
F18	SB/DEST 4	30					
F19	SB/DEST 3	32					
F20	SB/DEST 2	33					
F21	SB/DEST 1	34	2.4	-0.22	0.04	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F22	SB/DEST 0	35	2.4	-0.22	0.04	mA	1-3
F23	UIR BIT 0	36	2.4	-0.4	0.02	mA	1-3
F24	UIT BIT 1	37					
F25	DEAD OSC IN	39	2.4	-0.4	0.02	mA	1-3
F26	MSTR CLR	42	2.4	-0.2	0.02	mA	1-3
F27	UIR BIT 8	54	2.4	-2.0	0.05	mA	1-3
F28	RUN FF	55	2.4	-0.2	0.02	mA	1-3
F29	ADDR/WR 11	56	2.4	-0.5	0.02	mA	1-3
F30	ADDR/WR 0	57					
F31	ADDR/WR 1	58					
F32	ADDR/WR 2	59					
F33	ADDR/WR 3	61					
F34	ADDR/WR 4	62					
F35	ADDR/WR 5	63					
F36	ADDR/WR 6	64					
F37	ADDR/WR 7	65					
F38	ADDR/WR 8	66					
F39	ADDR/WR 9	67					
F40	ADDR/WR 10	68	2.4	-0.5	0.02	mA	1-3
F41	UIR BIT 2	69	2.4	-0.4	0.02	mA	1-3
F42	UIR BIT 3	70	2.4	-0.4	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F43	UIR BIT 4	71	2.4	-0.4	0.02	mA	1-3
F44	UIR BIT 5	72	2.4	-0.4	0.02	mA	1-3
F45	NDRO INIT	73	2.4	-2.7	0.09	mA	1-3
F46	UIR BIT 6	74	2.4	-0.2	0.02	mA	1-3
F47	S/D CLK	76	2.4	-8.4	0.35	mA	1-3
F48	BUS CLK	77	2.4	-4.0	0.07	mA	1-3
F49	UIR BIT 9	78	2.4	-0.2	0.02	mA	1-3
F50	UIR BIT 25	79	2.4	-0.2	0.02	mA	1-3
F51	UIR BIT 24	80	2.4	-0.4	0.02	mA	1-3
F52	EXT CLK IN	82					
F53	UIR BIT 26	83					
F54	UIR BIT 27	84					
F55	UIR BIT 11	85					
F56	UIR BIT 10	86					
F57	32 CLK SW	89					
F58	1 CLK SW	91					
F59	EXT CLK SW	92	2.4	-0.4	0.02	mA	1-3
F60	CMI PRES	93	2.4	-0.8	0.06	mA	1-3
F61	UIR BIT 13	94	2.4	-0.4	0.02	mA	1-3
F62	MC FMT	95					
F63	UIR BIT 14	96	2.4	-0.4	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F64	UIR BIT 12	97	2.4	-0.4	0.02	mA	1-3
F65	UIR BIT 15	98	2.4	-0.4	0.02	mA	1-3
F66	UIR BIT 28	99	2.4	-0.2	0.02	mA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F66
Power supply current limit	1.73 A	NA	F1-F66
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F66
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F66 F1-F66

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 31 uA	F1-F6
	NA	± 13 uA	F7-F22
	NA	± 21 uA	F23-F25
	NA	± 11 uA	F26
	NA	± 102 uA	F27
	NA	± 11 uA	F28
	NA	± 26 uA	F29-F40
	NA	± 21 uA	F41-F44
	NA	± 139 uA	F45
	NA	± 11 uA	F46
	NA	± 440 uA	F47
	NA	± 204 uA	F48
	NA	± 11 uA	F49,F50
	NA	± 21 uA	F51-F59
	NA	± 43 uA	F60
	NA	± 21 uA	F61-F65
	NA	± 11 uA	F66

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G1	PH SB ENBL	12	0.4	-0.6	0.02	mA	1-3
G2	UIR BIT 19	13					
G3	UIR BIT 20	14					
G4	UIR BIT 18	15					
G5	UIR BIT 17	16					
G6	UIR BIT 16	17	0.4	-0.6	0.02	mA	1-3
G7	SB/DEST 15	18	0.4	-0.22	0.04	mA	1-3
G8	SB/DEST 14	19					
G9	SB/DEST 13	20					
G10	SB/DEST 12	21					
G11	SB/DEST 11	22					
G12	SB/DEST 10	23					
G13	SB/DEST 9	24					
G14	SB/DEST 8	26					
G15	SB/DEST 7	27					
G16	SB/DEST 6	28					
G17	SB/DEST 5	29					
G18	SB/DEST 4	30					
G19	SB/DEST 3	32					
G20	SB/DEST 2	33					
G21	SB/DEST 1	34	0.4	-0.22	0.04	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G22	SB/DEST 0	35	0.4	-0.22	0.04	mA	1-3
G23	UIR BIT 0	36	0.4	-0.4	0.02	mA	1-3
G24	UIR BIT 1	37					
G25	DEAD OSC IN	39	0.4	-0.4	0.02	mA	1-3
G26	MSTR CLR	42	0.4	-0.2	0.02	mA	1-3
G27	UIR BIT 8	54	0.4	-2.0	0.05	mA	1-3
G28	RUN FF	55	0.4	-0.2	0.02	mA	1-3
G29	ADDR/WR 11	56	0.4	-0.5	0.02	mA	1-3
G30	ADDR/WR 0	57					
G31	ADDR/WR 1	58					
G32	ADDR/WR 2	59					
G33	ADDR/WR 3	61					
G34	ADDR/WR 4	62					
G35	ADDR/WR 5	63					
G36	ADDR/WR 6	64					
G37	ADDR/WR 7	65					
G38	ADDR/WR 8	66					
G39	ADDR/WR 9	67					
G40	ADDR/WR 10	68	0.4	-0.5	0.02	mA	1-3
G41	UIR BIT 2	69	0.4	-0.4	0.02	mA	1-3
G42	UIR BIT 3	70	0.4	-0.4	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G43	UIR BIT 4	71	0.4	-0.4	0.02	mA	1-3
G44	UIR BIT 5	72	0.4	-0.4	0.02	mA	1-3
G45	NDRO INIT	73	0.4	-2.7	0.09	mA	1-3
G46	UIR BIT 6	74	0.4	-0.2	0.02	mA	1-3
G47	S/D CLK	76	0.4	-8.4	0.35	mA	1-3
G48	BUS CLK	77	0.4	-4.0	0.07	mA	1-3
G49	UIR BIT 9	78	0.4	-0.2	0.02	mA	1-3
G50	UIR BIT 25	79	0.4	-0.2	0.02	mA	1-3
G51	UIR BIT 24	80	0.4	-0.4	0.02	mA	1-3
G52	EXT CLK IN	82					
G53	UIR BIT 26	83					
G54	UIR BIT 27	84					
G55	UIR BIT 11	85					
G56	UIR BIT 10	86					
G57	32 CLK SW	89					
G58	1 CLK SW	91					
G59	EXT CLK SW	92	0.4	-0.4	0.02	mA	1-3
G60	CMI PRES	93	0.4	-0.8	0.06	mA	1-3
G61	UIR BIT 13	94	0.4	-0.4	0.02	mA	1-3
G62	MC FMT	95					
G63	UIR BIT 14	96	0.4	-0.4	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G64	UIR BIT 12	97	0.4	-0.4	0.02	mA	1-3
G65	UIR BIT 15	98	0.4	-0.4	0.02	mA	1-3
G66	UIR BIT 28	99	0.4	-0.2	0.02	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G66
Power supply current limit	1.73 A	NA	G1-G66
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G66
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 G1-G66 G1-G66
Input current measurement	NA NA NA NA NA NA NA NA	± 31 uA ± 13 uA ± 21 uA ± 11 uA ± 102 uA ± 11 uA ± 26 uA ± 21 uA	G1-G6 G7-G22 G23-G25 G26 G27 G28 G29-G40 G41-G44

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 139 uA	G45
	NA	± 11 uA	G46
	NA	± 440 uA	G47
	NA	± 204 uA	G48
	NA	± 11 uA	G49,G50
	NA	± 21 uA	G51-G59
	NA	± 43 uA	G60
	NA	± 21 uA	G61-G65
	NA	± 11 uA	G66

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	MEM RD 0	2	-0.4	2.4	4.5	V	1-4
H2	MEM RD 1	3					
H3	MEM RD 2	4					
H4	MEM RD 3	5					
H5	MEM RD 4	6					
H6	MEM RD 5	7					
H7	MEM RD 6	8					
H8	MEM RD 7	9	-0.4	2.4	4.5	V	1-4
H9	SB/DEST 15	18	-0.16	2.4	4.5	V	1-4
H10	SB/DEST 14	19					
H11	SB/DEST 13	20					
H12	SB/DEST 12	21					
H13	SB/DEST 11	22					
H14	SB/DEST 10	23					
H15	SB/DEST 9	24					
H16	SB/DEST 8	26					
H17	SB/DEST 7	27					
H18	SB/DEST 6	28					
H19	SB/DEST 5	29					
H20	SB/DEST 4	30					
H21	SB/DEST 3	32	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H22	SB/DEST 2	33	-0.16	2.4	4.5	V	1-4
H23	SB/DEST 1	34					
H24	SB/DEST 0	35					
H25	DDSTK INTR	38	-0.16	2.4	4.5	V	1-4
H26	MEM RD 8	43	-0.4	2.4	4.5	V	1-4
H27	MEM RD 9	44					
H28	MEM RD 10	45					
H29	MEM RD 11	46					
H30	MEM RD 12	47					
H31	MEM RD 13	48					
H32	MEM RD 14	49					
H33	MEM RD 15	50					
H34	MEM BSY	52					
H35	DATA AVAIL	53	-0.4	2.4	4.5	V	1-4
H36	MON CLK INTR	87	-0.16	2.4	4.5	V	1-4
H37	RTC INTR	88					
H38	1 OR 32 KHZ	100	-0.16	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H38
Power supply current limit	1.73 A	NA	H1-H38
Forcing current, (Ioh)	-0.4 mA -0.16 mA -0.4 mA -0.16 mA	± 40 uA ± 16 uA ± 40 uA ± 16 uA	H1-H8 H9-H25 H26-H35 H36-H38
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 H1-H38 H1-H38
Output voltage measurement	NA	± 105 mV	H1-H38

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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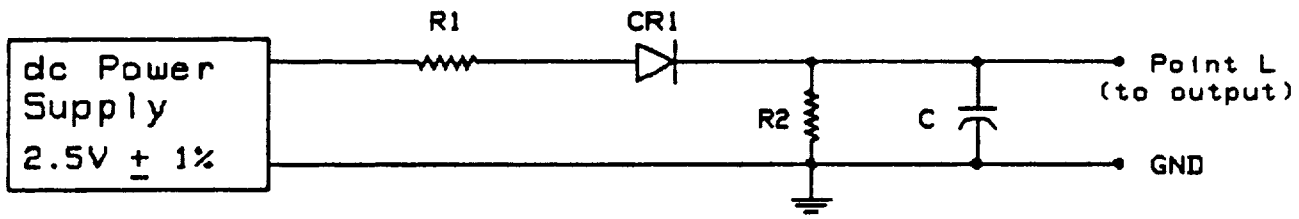
NOTES: - Continued.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

    (a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

    (b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

    (c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

    (d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
2,3,4,5,6,7,8,9, 43,44,45,46,47, 48,49,50,52,53	6200	6.0	0.5
18,19,20,21,22, 23,24,26,27,28, 29,30,32,33,34, 35,38,87,88,100	16000	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	MEM RD 0	2	6.0	0.0	0.5	V	1-4
J2	MEM RD 1	3					
J3	MEM RD 2	4					
J4	MEM RD 3	5					
J5	MEM RD 4	6					
J6	MEM RD 5	7					
J7	MEM RD 6	8					
J8	MEM RD 7	9	6.0	0.0	0.5	V	1-4
J9	SB/DEST 15	18	1.6	0.0	0.5	V	1-4
J10	SB/DEST 14	19					
J11	SB/DEST 13	20					
J12	SB/DEST 12	21					
J13	SB/DEST 11	22					
J14	SB/DEST 10	23					
J15	SB/DEST 9	24					
J16	SB/DEST 8	26					
J17	SB/DEST 7	27					
J18	SB/DEST 6	28					
J19	SB/DEST 5	29					
J20	SB/DEST 4	30					
J21	SB/DEST 3	32	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J22	SB/DEST 2	33	1.6	0.0	0.5	V	1-4
J23	SB/DEST 1	34					
J24	SB/DEST 0	35					
J25	DDSTK INTR	38	1.6	0.0	0.5	V	1-4
J26	MEM RD 8	43	6.0	0.0	0.5	V	1-4
J27	MEM RD 9	44					
J28	MEM RD 10	45					
J29	MEM RD 11	46					
J30	MEM RD 12	47					
J31	MEM RD 13	48					
J32	MEM RD 14	49					
J33	MEM RD 15	50					
J34	MEM BSY	52					
J35	DATA AVAIL	53	6.0	0.0	0.5	V	1-4
J36	MON CLK INTR	87	1.6	0.0	0.5	V	1-4
J37	RTC INTR	88					
J38	1 OR 32 KHZ	100	1.6	0.0	0.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J38
Power supply current limit	1.73 A	NA	J1-J38
Forcing current, (Iol)	6.0 mA 1.6 mA 6.0 mA 1.6 mA	± 600 uA ± 160 uA ± 600 uA ± 160 uA	J1-J8 J9-J25 J26-J35 J36-J38
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 J1-J38 J1-J38
Output voltage measurement	NA	± 25 mV	J1-J38

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
High level							
K1	MEM RD 0	2	2.4	-0.01	0.01	mA	1-3
K2	MEM RD 1	3					
K3	MEM RD 2	4					
K4	MEM RD 3	5					
K5	MEM RD 4	6					
K6	MEM RD 5	7					
K7	MEM RD 6	8					
K8	MEM RD 7	9					
K9	MEM RD 8	43					
K10	MEM RD 9	44					
K11	MEM RD 10	45					
K12	MEM RD 11	46					
K13	MEM RD 12	47					
K14	MEM RD 13	48					
K15	MEM RD 14	49					
K16	MEM RD 15	50	2.4	-0.01	0.01	mA	1-3
K17	MEM BUSY	52	2.4	-0.05	0.05	mA	1-3
K18	DATA AVAIL	53	2.4	-0.05	0.05	mA	1-3
Low level							
K19	MEM RD 0	2	0.4	-0.01	0.01	mA	1-3

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TABLE XI. Output leakage current, high and low level, test - Continued.

Test cond	Signal name	Pin No.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K20	MEM RD 1	3	0.4	-0.01	0.01	mA	1-3
K21	MEM RD 2	4					
K22	MEM RD 3	5					
K23	MEM RD 4	6					
K24	MEM RD 5	7					
K25	MEM RD 6	8					
K26	MEM RD 7	9					
K27	MEM RD 8	43					
K28	MEM RD 9	44					
K29	MEM RD 10	45					
K30	MEM RD 11	46					
K31	MEM RD 12	47					
K32	MEM RD 13	48					
K33	MEM RD 14	49					
K34	MEM RD 15	50	0.4	-0.01	0.01	mA	1-3
K35	MEM BUSY	52	0.4	-0.05	0.05	mA	1-3
K36	DATA AVAIL	53	0.4	-0.05	0.05	mA	1-3

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NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K36
Power supply current limit	1.73 A	NA	K1-K36
Forcing voltage, (Voh) (Vol)	 2.4 V 0.4 V	 ± 50 mV ± 50 mV	 K1-K18 K19-K36
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 K1-K36 K1-K36
Output current measurement	NA NA NA NA	± 1 uA ± 5 uA ± 1 uA ± 5 uA	K1-K16 K17,K18 K19-K34 K35,K36

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	PH SB ENBL	12	-18.0	-1.2	0.0	V	1-3
L2	UIR BIT 19	13					
L3	UIR BIT 20	14					
L4	UIR BIT 18	15					
L5	UIR BIT 17	16					
L6	UIR BIT 16	17	-18.0	-1.2	0.0	V	1-3
L7	UIR BIT 0	36	-18.0	-1.5	0.0	V	1-3
L8	UIR BIT 1	37					
L9	DEAD OSC IN	39					
L10	MSTR CLR	42	-18.0	-1.5	0.0	V	1-3
L11	UIR BIT 8	54	-18.0	-1.2	0.0	V	1-3
L12	RUN FF	55	-18.0	-1.5	0.0	V	1-3
L13	ADDR/WR 11	56					
L14	ADDR/WR 0	57					
L15	ADDR/WR 1	58					
L16	ADDR/WR 2	59					
L17	ADDR/WR 3	61					
L18	ADDR/WR 4	62					
L19	ADDR/WR 5	63					
L20	ADDR/WR 6	64					
L21	ADDR/WR 7	65	-18.0	-1.5	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L22	ADDR/WR 8	66	-18.0	-1.5	0.0	V	1-3
L23	ADDR/WR 9	67					
L24	ADDR/WR 10	68					
L25	UIR BIT 2	69					
L26	UIR BIT 3	70					
L27	UIR BIT 4	71					
L28	UIR BIT 5	72					
L29	NDRO INIT	73					
L30	UIR BIT 6	74					
L31	S/D CLK	76	-18.0	-1.5	0.0	V	1-3
L32	BUS CLK	77	-18.0	-1.2	0.0	V	1-3
L33	UIR BIT 9	78	-18.0	-1.5	0.0	V	1-3
L34	UIR BIT 25	79					
L35	UIR BIT 24	80					
L36	EXT CLK IN	82					
L37	UIR BIT 26	83					
L38	UIR BIT 27	84					
L39	UIR BIT 11	85					
L40	UIR BIT 10	86					
L41	32 CLK SW	89					
L42	1 CLK SW	91	-18.0	-1.5	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L43	EXT CLK SW	92	-18.0	-1.5	0.0	V	1-3
L44	CMI PRES	93					
L45	UIR BIT 13	94					
L46	MC FMT	95					
L47	UIR BIT 14	96					
L48	UIR BIT 12	97					
L49	UIR BIT 15	98					
L50	UIR BIT 28	99	-18.0	-1.5	0.0	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L50
Power supply current limit	NA	NA	L1-L50
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L50

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L50 L1-L50
Input voltage measurement	NA NA NA NA NA NA	± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV	L1-L6 L7-L10 L11 L12-L31 L32 L33-L50

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	4.9	9.1	uF	1-4
M2	PSFC	51	7.0	13.0	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 210 nF ± 300 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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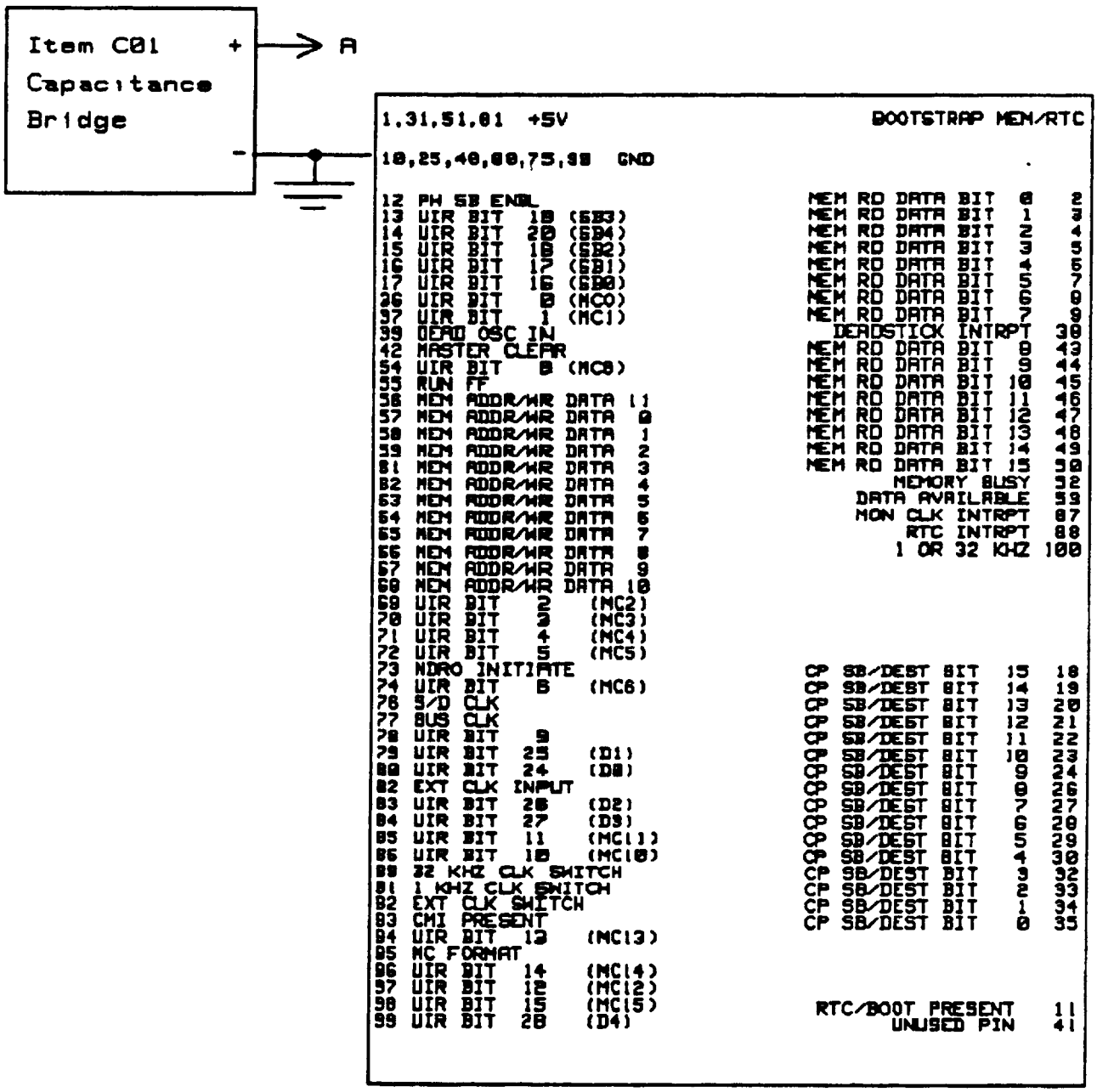


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	480	1150	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	1.73 A	NA	N
Pattern voltage high, (VH)	3.0 V	± 100 mV	N
low, (VL)	0.0 V	± 100 mV	N
Power supply current measurement	NA	± 33.5 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (250 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (250 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	1.73 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 25 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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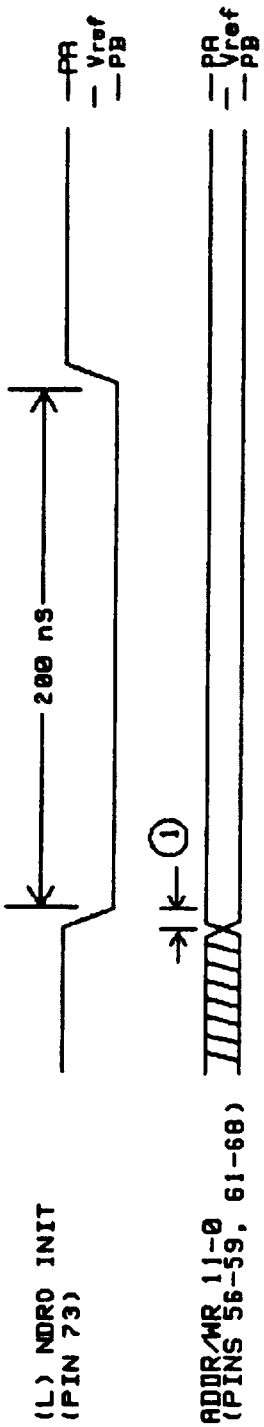


FIGURE 11. Input timing parameters for table XV.

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NOTE:

1. (1) ADDR/WR 0,ADDR/WR 1,ADDR/WR 2,ADDR/WR 3,ADDR/WR 4, ADDR/WR 5,ADDR/WR 6,ADDR/WR 7,ADDR/WR 8,ADDR/WR 9,ADDR/WR 10,ADDR/WR 11 TO NDRO INIT. Setup time: minimum 5 nS.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (250 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (250 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (250 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (250 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	1.73 A 1.73 A	Q1,Q3 Q2,Q4
Input voltage high, (VH) low, (VL)	2.4 V 0.4 V	± 100 mV ± 100 mV	NA NA	Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Current limit	Test condition
Output voltage verification				
high, (Voh)	NA	± 100 mV	NA	Q1,Q3
high, (Voh)	NA	± 100 mV	NA	Q2,Q4
low, (Vol)	NA	± 100 mV	NA	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

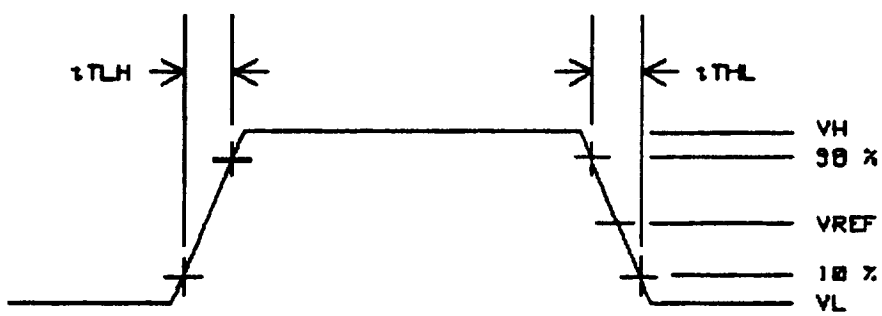
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tsu	ADDR/WR 11 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 0 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 1 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 2 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 3 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 4 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 5 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 6 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 7 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 8 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 9 to NDRO INIT	5	5	5	-	-	-	nS	12
1	tsu	ADDR/WR 10 to NDRO INIT	5	5	5	-	-	-	nS	12
2	tpd	BUS CLK to MEM BSY	-	-	-	38	38	38	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	NDRO INIT to MEM RD 0	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 1	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 2	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 3	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 4	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 5	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 6	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 7	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 8	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 9	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 10	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 11	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 12	-	-	-	109	109	109	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	NDRO INIT to MEM RD 13	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 14	-	-	-	109	109	109	nS	12
3	tpd	NDRO INIT to MEM RD 15	-	-	-	109	109	109	nS	12
4	tpd	UIR BIT 19 to SB/DEST 15	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 14	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 13	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 12	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 11	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 10	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 9	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 8	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 7	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 6	-	-	-	59	59	59	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	UIR BIT 19 to SB/DEST 5	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 4	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 3	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 2	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 1	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 19 to SB/DEST 0	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 15	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 14	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 13	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 12	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 11	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 10	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 9	-	-	-	59	59	59	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	UIR BIT 20 to SB/DEST 8	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 7	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 6	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 5	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 4	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 3	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 2	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 1	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 20 to SB/DEST 0	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 15	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 14	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 13	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 12	-	-	-	59	59	59	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	UIR BIT 18 to SB/DEST 11	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 10	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 9	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 8	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 7	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 6	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 5	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 4	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 3	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 2	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 1	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 18 to SB/DEST 0	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 15	-	-	-	59	59	59	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	UIR BIT 17 to SB/DEST 14	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 13	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 12	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 11	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 10	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 9	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 8	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 7	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 6	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 5	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 4	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 3	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 2	-	-	-	59	59	59	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	UIR BIT 17 to SB/DEST 1	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 17 to SB/DEST 0	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 15	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 14	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 13	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 12	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 11	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 10	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 9	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 8	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 7	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 6	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 5	-	-	-	59	59	59	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	UIR BIT 16 to SB/DEST 4	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 3	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 2	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 1	-	-	-	59	59	59	nS	12
4	tpd	UIR BIT 16 to SB/DEST 0	-	-	-	59	59	59	nS	12
5	tpd	S/D CLK to MON CLK INTR	-	-	-	53	53	53	nS	12
5	tpd	S/D CLK to RTC INTR	-	-	-	53	53	53	nS	12
6	tpd	DEAD OSC IN to DDSTK INTR	-	-	-	80	80	80	nS	12
7	tpd	BUS CLK to DATA AVAIL	-	-	-	52	52	52	nS	12

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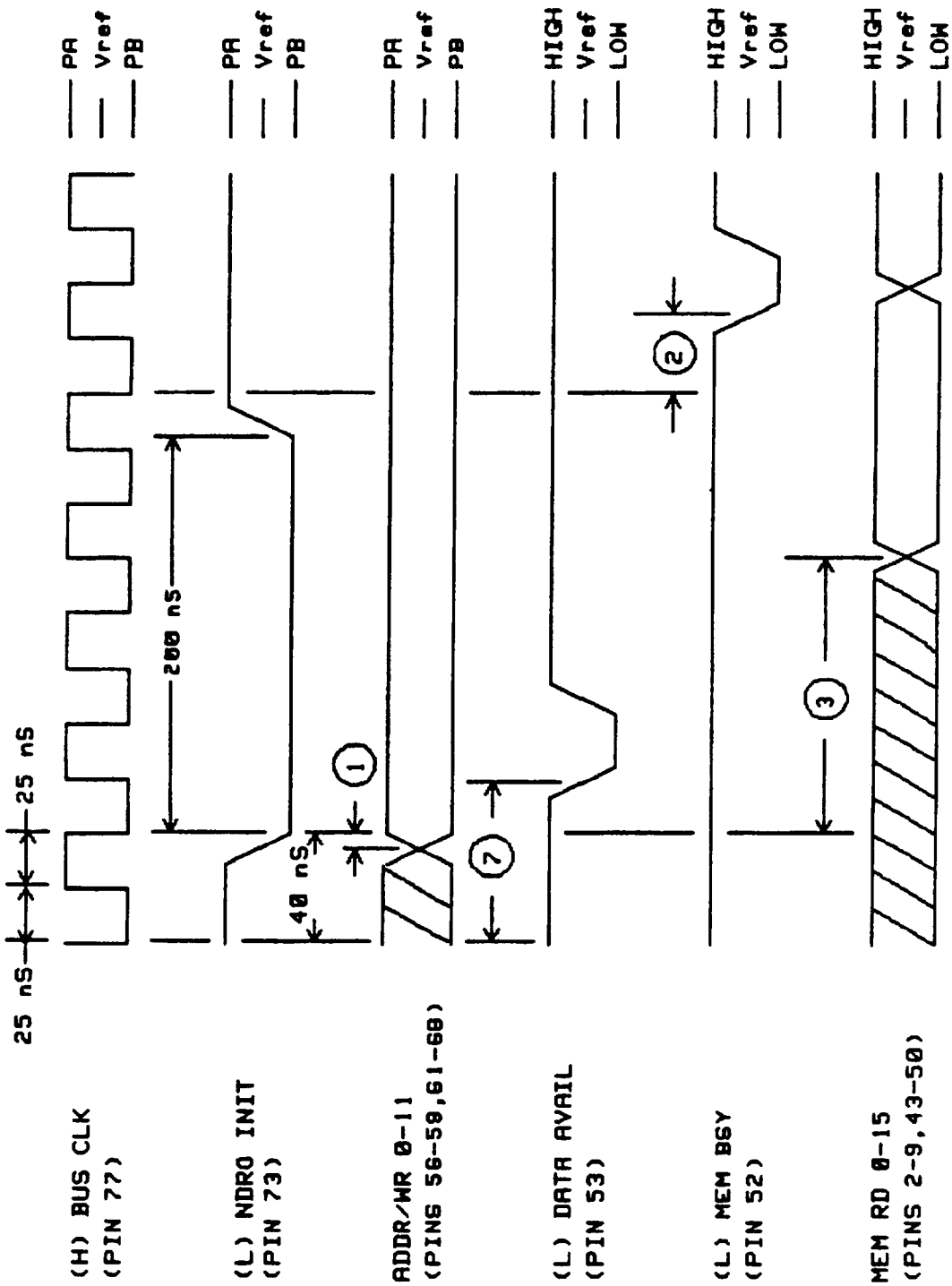


FIGURE 12a. Dynamic waveforms for table XVII.

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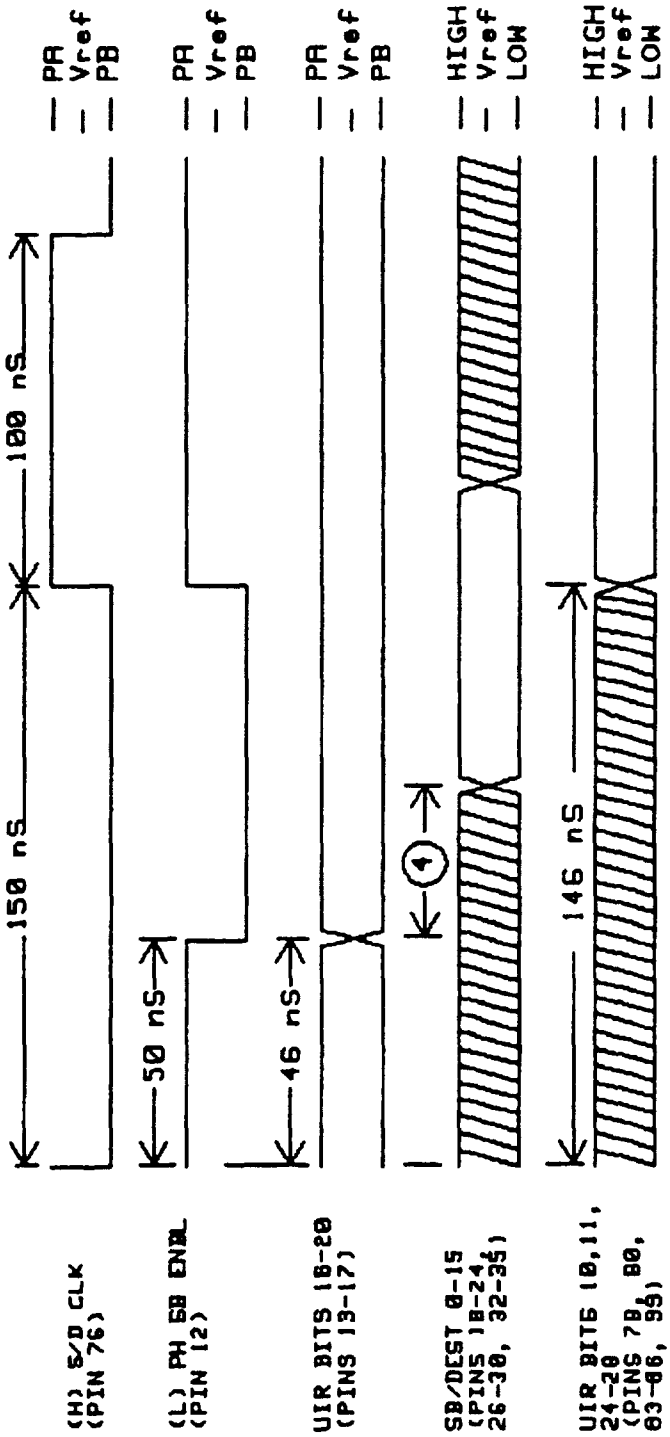
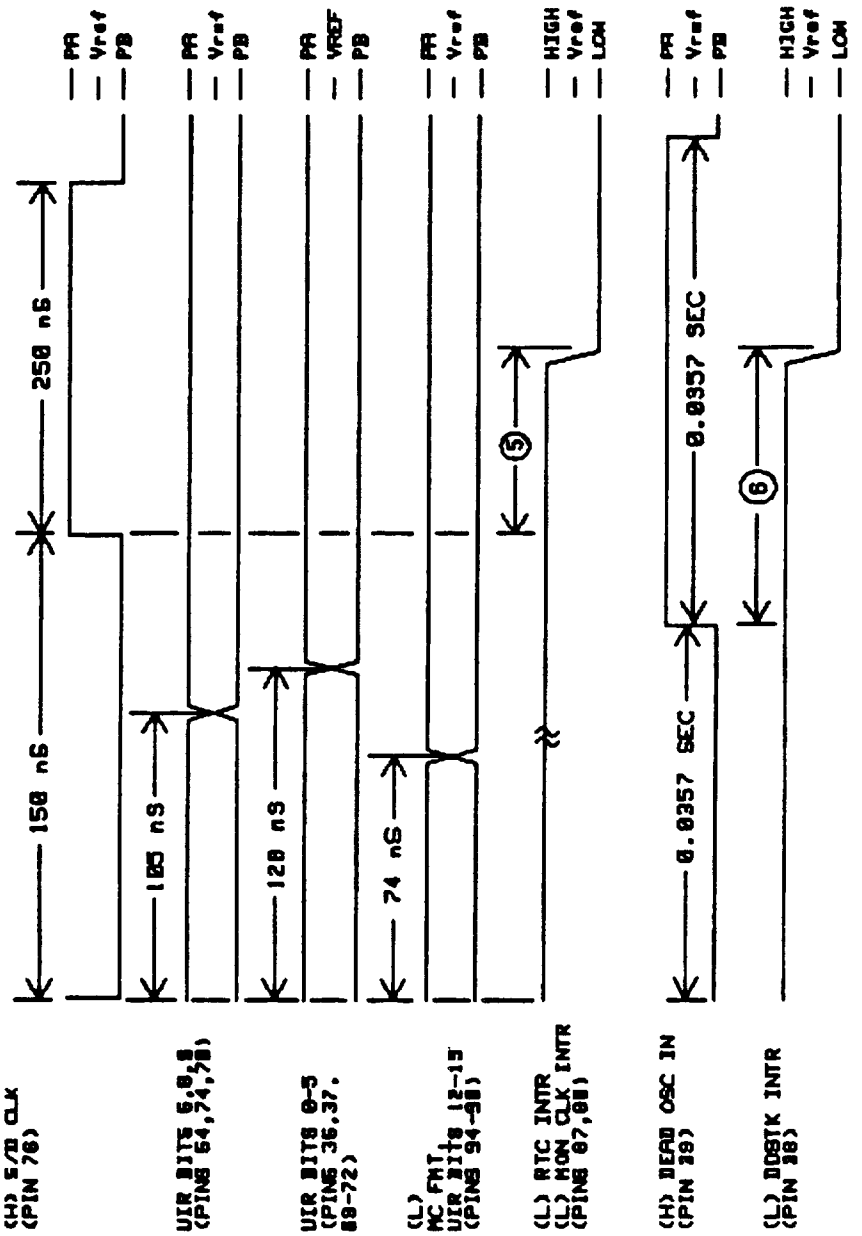


FIGURE 12b. Dynamic waveforms for table XVII.

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NOTE:

1. The deadstick oscillator is totally asynchronous to any other clock.

FIGURE 12c. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	PH SB ENBL	12	2	50	pF	1-4
R2	UIR BIT 19	13				
R3	UIR BIT 20	14				
R4	UIR BIT 18	15				
R5	UIR BIT 17	16				
R6	UIR BIT 16	17	2	50	pF	1-4
R7	UIR BIT 0	36	2	75	pF	1-4
R8	UIR BIT 1	37				
R9	DEAD OSC IN	39				
R10	MSTR CLR	42				
R11	UIR BIT 8	54	2	75	pF	1-4
R12	RUN FF	55	2	50	pF	1-4
R13	ADDR/WR 11	56	4	125	pF	1-4
R14	ADDR/WR 0	57	4	100	pF	1-4
R15	ADDR/WR 1	58	4	125	pF	1-4
R16	ADDR/WR 2	59	4	125	pF	1-4
R17	ADDR/WR 3	61	4	100	pF	1-4
R18	ADDR/WR 4	62	4	125	pF	1-4
R19	ADDR/WR 5	63				
R20	ADDR/WR 6	64	4	125	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	ADDR/WR 7	65	4	100	pF	1-4
R22	ADDR/WR 8	66	4	125	pF	1-4
R23	ADDR/WR 9	67				
R24	ADDR/WR 10	68	4	125	pF	1-4
R25	UIR BIT 2	69	2	50	pF	1-4
R26	UIR BIT 3	70				
R27	UIR BIT 4	71				
R28	UIR BIT 5	72	2	50	pF	1-4
R29	NDRO INIT	73	8	150	pF	1-4
R30	UIR BIT 6	74	2	75	pF	1-4
R31	S/D CLK	76	2	175	pF	1-4
R32	BUS CLK	77	4	75	pF	1-4
R33	UIR BIT 9	78	2	75	pF	1-4
R34	UIR BIT 25	79	2	50	pF	1-4
R35	UIR BIT 24	80				
R36	EXT CLK IN	82				
R37	UIR BIT 26	83				
R38	UIR BIT 27	84				
R39	UIR BIT 11	85				
R40	UIR BIT 10	86				
R41	32 CLK SW	89	2	50	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R42	1 CLK SW	91	2	50	pF	1-4
R43	EXT CLK SW	92	2	50	pF	1-4
R44	CMI PRES	93	6	100	pF	1-4
R45	UIR BIT 13	94	2	50	pF	1-4
R46	MC FMT	95				
R47	UIR BIT 14	96				
R48	UIR BIT 12	97				
R49	UIR BIT 15	98	2	50	pF	1-4
R50	UIR BIT 28	99	2	75	pF	1-4
Outputs						
R51	MEM RD 0	2	2	50	pF	1-4
R52	MEM RD 1	3				
R53	MEM RD 2	4				
R54	MEM RD 3	5	2	50	pF	1-4
R55	MEM RD 4	6	2	75	pF	1-4
R56	MEM RD 5	7				
R57	MEM RD 6	8				
R58	MEM RD 7	9	2	75	pF	1-4
R59	MEM RD 8	43	2	50	pF	1-4
R60	MEM RD 9	44				
R61	MEM RD 10	45	2	50	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R62	MEM RD 11	46	2	50	pF	1-4
R63	MEM RD 12	47				
R64	MEM RD 13	48				
R65	MEM RD 14	49	2	50	pF	1-4
R66	MEM RD 15	50	2	75	pF	1-4
Bidirectionals						
R67	SB/DEST 15	18	4	75	pF	1-4
R68	SB/DEST 14	19				
R69	SB/DEST 13	20				
R70	SB/DEST 12	21				
R71	SB/DEST 11	22				
R72	SB/DEST 10	23				
R73	SB/DEST 9	24				
R74	SB/DEST 8	26				
R75	SB/DEST 7	27				
R76	SB/DEST 6	28				
R77	SB/DEST 5	29				
R78	SB/DEST 4	30				
R79	SB/DEST 3	32	4	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
R80	SB/DEST 2	33	4	75	pF	1-4
R81	SB/DEST 1	34				
R82	SB/DEST 0	35	4	75	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R82
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R82
Capacitance	NA	± 2.4 pF	R1-R6
	NA	± 3.65 pF	R7-R11
	NA	± 2.4 pF	R12
	NA	± 6.05 pF	R13
	NA	± 4.8 pF	R14
	NA	± 6.05 pF	R15,R16
	NA	± 4.8 pF	R17
	NA	± 6.05 pF	R18-R20
	NA	± 4.8 pF	R21
	NA	± 6.05 pF	R22-R24
	NA	± 2.4 pF	R25-R28
	NA	± 7.1 pF	R29
	NA	± 3.65 pF	R30
	NA	± 8.65 pF	R31
	NA	± 3.55 pF	R32
	NA	± 3.65 pF	R33
	NA	± 2.4 pF	R34-R43
	NA	± 4.7 pF	R44
	NA	± 2.4 pF	R45-R49
	NA	± 3.65 pF	R50
	NA	± 2.4 pF	R51-R54
	NA	± 3.65 pF	R55-R58
	NA	± 2.4 pF	R59-R65
	NA	± 3.65 pF	R66
	NA	± 3.55 pF	R67-R82

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NOTES: - Continued.

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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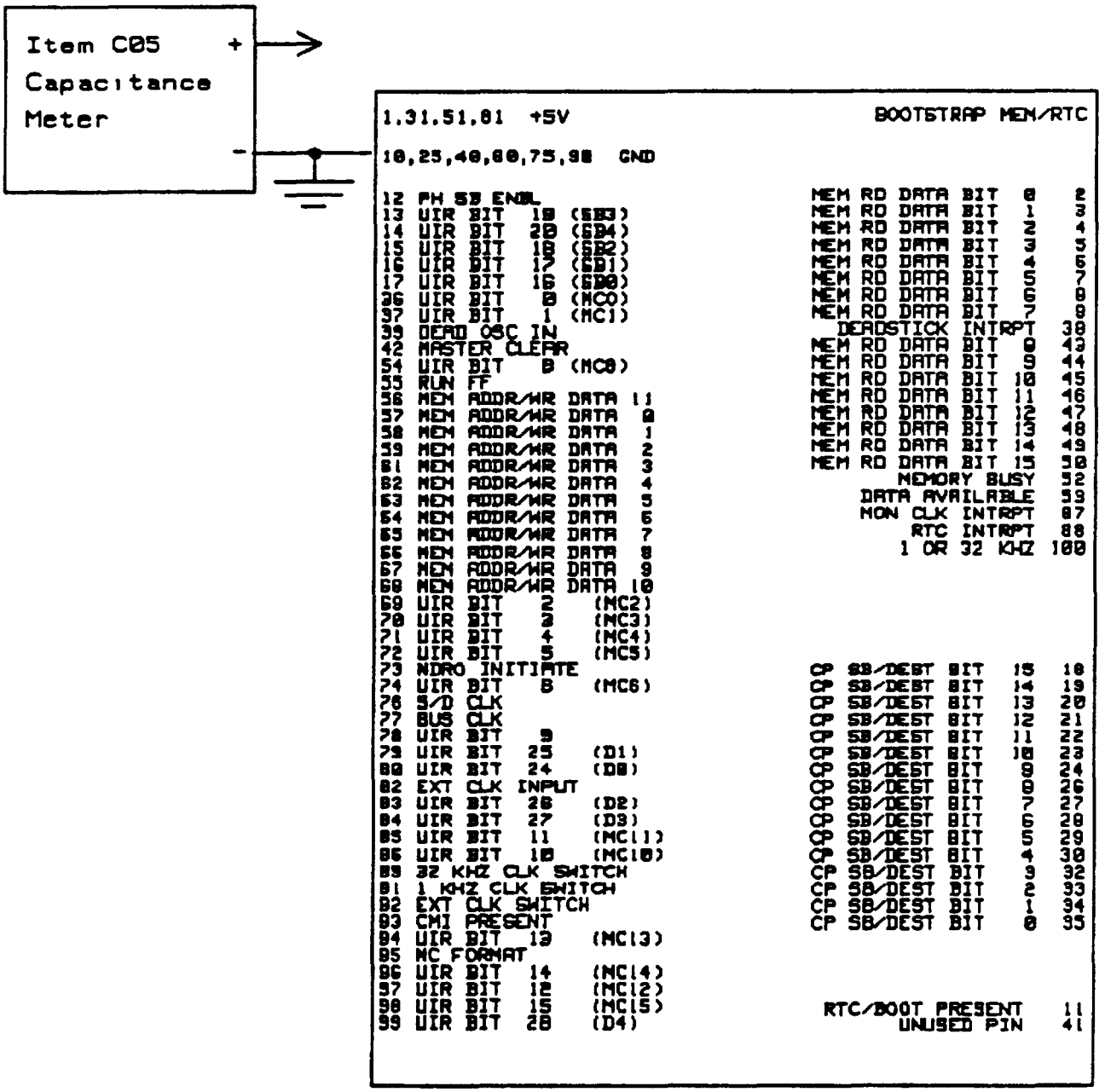


FIGURE 13. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA	1	0	400	mV	1-4
T2	PSTA	51	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	1.73 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as indicated on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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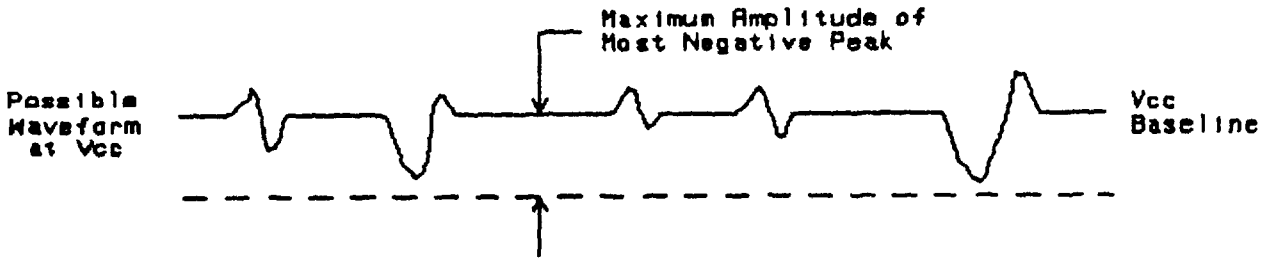
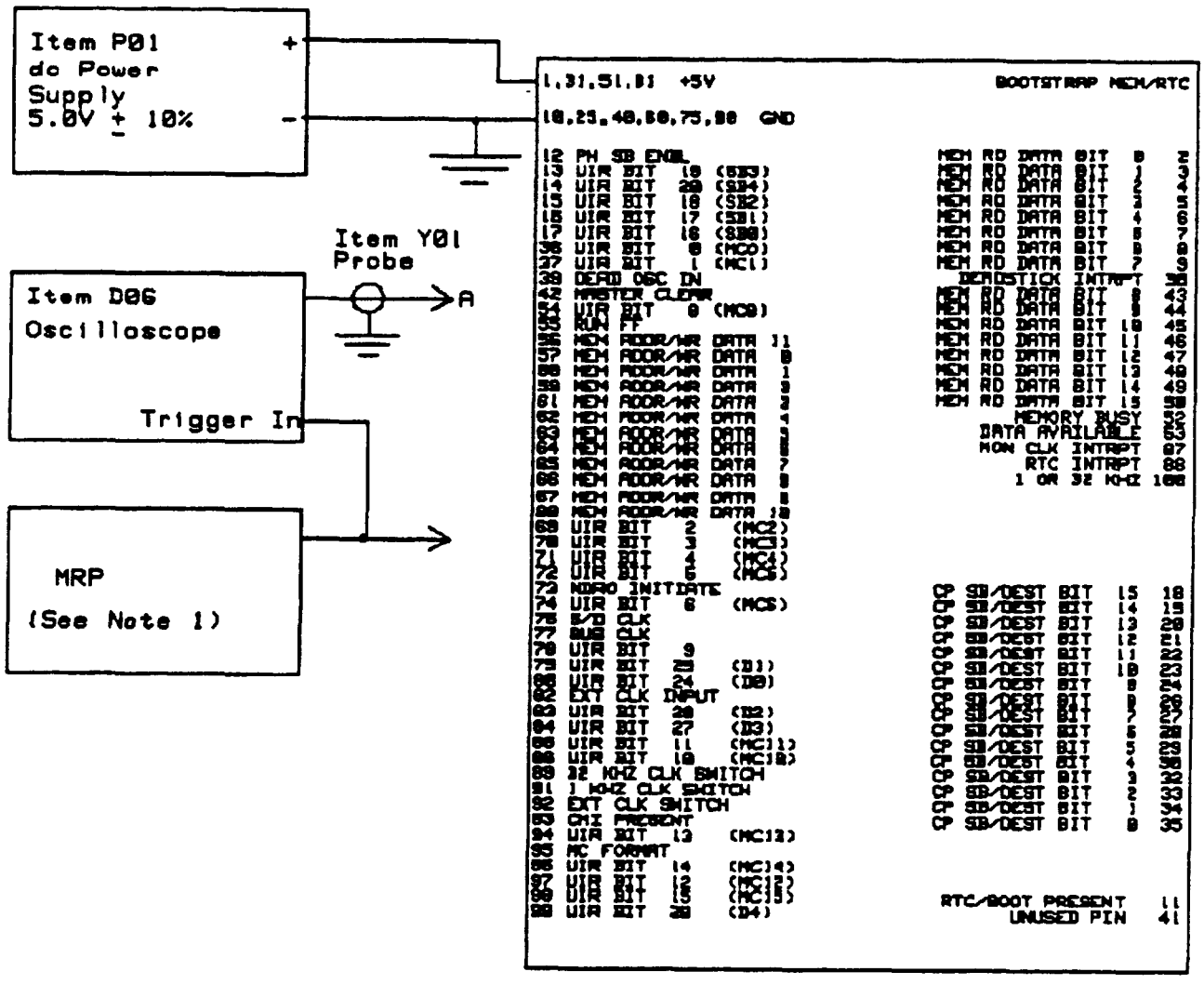


FIGURE 14. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XIX.

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NAVAL SEA SYSTEMS COMMAND (SEA 5523)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/518	Modules, Standard Electronic, Bootstrap Memory/RTC Module, Key Code CWAF

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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